Structure, Bonding, and Self-Assembly of Biological Structures

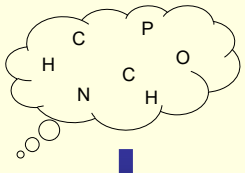
CHEM0010/1 – Part A

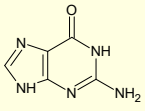
Dr. Anna Roffey
Academic Support Hours:
Mon 11:30-12:30
Thurs 2-3 pm (CIB 124h)

Lecture 1

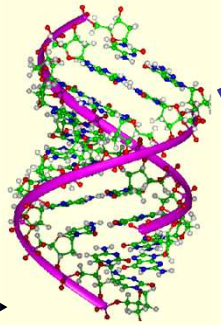


This course is about how atoms bond to give molecules, and how molecules combine into multi-molecular structures:





Guanine



Double-stranded DNA



Injection of virus DNA into a liposome

Learning Objectives: Atomic Structure and Covalent Bonding

1. ATOMIC STRUCTURE: To recall the **electron configuration of atoms**, and how patterns of electron configuration are represented in the periodic table.

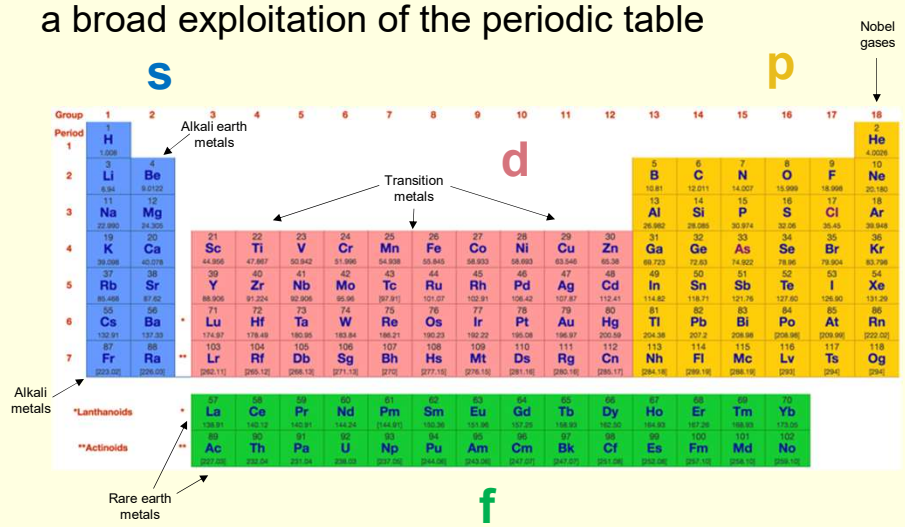
2. To understand and predict the COVALENT BONDING in simple molecules made up of C, O, H, N, S, P using **electron-sharing** arguments (Lewis structures).

3. To understand the geometry of covalently bonded molecules based on **valence shell electron pair repulsion** (VSEPR).

4. To introduce **orbital overlap** and **hybridisation** models for bonding (Valence Bond Theory). To draw single, double and triple bonds, and rationalise these using diagrams for orbital overlap.

5. To introduce basic **Molecular Orbital Theory**, as an alternative (and sometimes more accurate) representation of bonding in molecules, and apply this to diatomic molecules.

Nature's tool-box: a broad exploitation of the periodic table



Elemental composition of the human body

O : most abundant, at 65% your mass, largely in water but also in all proteins, carbohydrates, lipids, DNA.

C : around 18% of your mass, the principal ingredient in all proteins, sugars, lipids, DNA.

H : around 10% of your mass, but that means more atoms of H than of Oxygen or Carbon!

N : another principal ingredient, at 3%, included in all protein, DNA, and many other molecules.

Ca : 1.5% of the human mass, responsible for signalling and osmotic functions

P : 1.2 %. Phosphates, including mono- , di- and tri- phosphates which are crucial in the energy transfer.

K : around 0.2% of each human, in K+ form as electrolyte in plasma. Osmotic function.

Na : around 0.1% of each human, in Na+ form as electrolyte in plasma. Osmotic function.

S : 0.2% . Amino acid Cysteine

Cl : 0.2%

Mg : 0.05%. In many enzymes, and essential to all cells, as chelator of ATP

Fe : At the centre of Haem group in Haemoglobin

Co : 'metal centre' of vitamin B_{12}

Cu : trace, e.g. in cytochrome C oxidase enzyme.

Zn : reactive centre in many enzymes, e.g. alcohol dehydrogenase

I : ingredient of some thyroid hormones

Se :

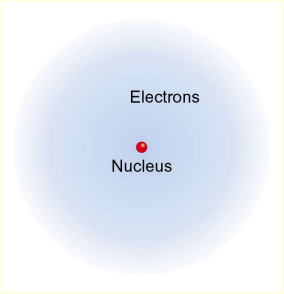
F :

Etc etc etc...

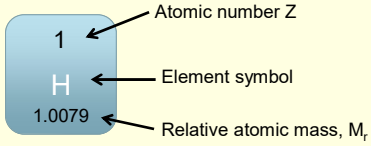
Atomic structure

Atoms are composed of subatomic particles:

	mass	*charge*	*spin*
Electrons	0	-1	½
Protons	1	+1	½
Neutrons	1	0	½



The nucleus is composed of protons and neutrons and the different elements are identified by their **Atomic number Z**, which indicates the number of protons in the nucleus.



The composition of the nucleus determines the **mass** of an element

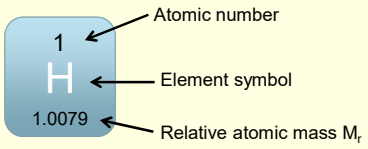
For a given element the number of **protons** is fixed, while the number of **neutrons** can vary.

Atoms of the same element but with differing numbers of neutrons (and therefore different mass) are called **isotopes**.

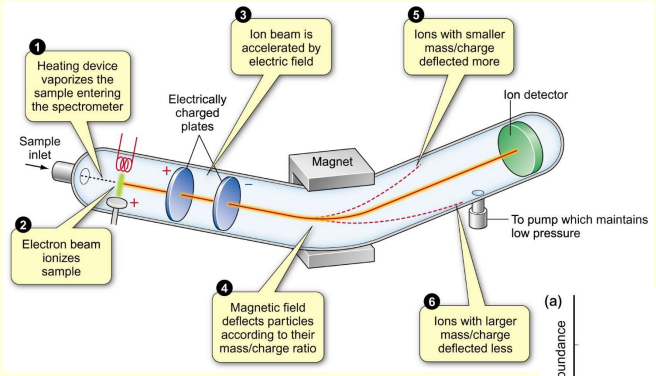
The isotopes of hydrogen with 0, 1, and 2 neutrons are written as:

Atomic mass → 1
Atomic Number → 1 $^{1}_{1}H$ $^{2}_{1}H$ $^{3}_{1}H$

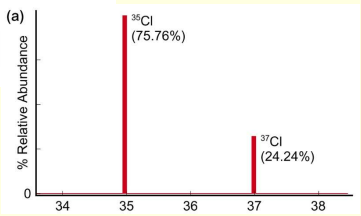
The relative atomic mass M_r of an element, as shown on a periodic table, takes account of the natural abundance of each isotope of an element. Therefore, the values are not necessarily integers. For example:

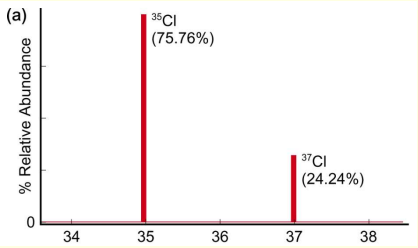


Using a **mass spectrometer**, the mass of the individual isotopes can be determined.



e.g. for Cl which has two different isotopes of mass 35 and 37



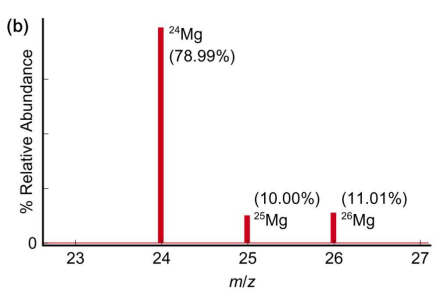


Cl has two different isotopes of mass 35 and 37

Relative atomic mass of naturally occurring chlorine

$$= \frac{(35 \times 75.76) + (37 \times 24.24)}{100}$$

= 35.45 u



Mg has three different isotopes of mass 24 and 25 and 26

Relative atomic mass of naturally occurring magnesium

$$= \frac{(24 \times 78.99) + (25 \times 10.00) + (26 \times 11.01)}{100}$$

= 24.31 u

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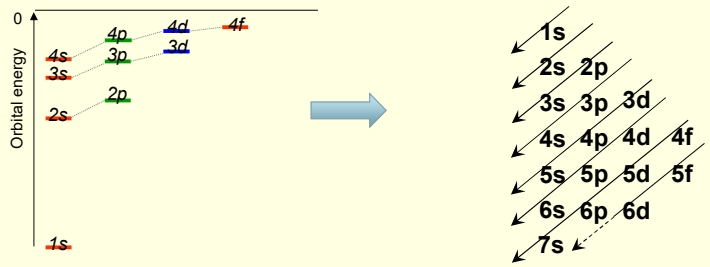
Electrons

So the nucleus is important in determining the mass of an atom, but chemical reactivity is largely down to **electronic configuration,** i.e. how many electrons an atom has and how they are arranged around the nucleus.

From A level you will know that electrons occupy **orbitals** that have **fixed (discrete)** energy values.

There are rules about which order the orbitals are occupied and how many electrons may occupy each orbital.

Lowest energy orbitals are occupied first (**Aufbau 'building up' principle**) and only two electrons may occupy each orbital (**Pauli principle**)



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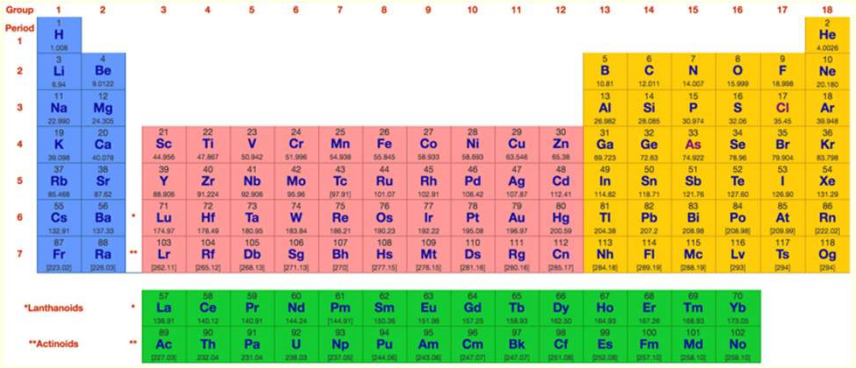
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e.g. Helium Z = 2 $1s^2$

Carbon Z = 6 $1s^2\ 2s^2\ 2p^2$

Potassium Z = 19 $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$ or [Ar] $4s^1$

Zirconium Z = 40 [Kr] $5s^2\ 4d^2$



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How do we know the relative energies of the orbitals and which are occupied?

From **experimental** observation of **atomic spectroscopy**. These experiments involve the excitation of electrons between the different orbitals and the determination of which transitions are observed.

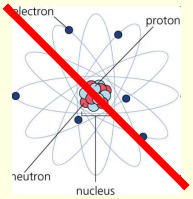
You will learn about this in more detail in Section B of the course.

Why are the orbitals arranged in this order and why are only two electrons found in each orbital?

OK, so now we are going to have to delve a little into the world of **quantum mechanics** and first we are going to talk more about the nature of the electron and what is an orbital….?

Electrons and Orbitals

In some ways its easiest to start off with what an electron is not. It's not a discreet particle with an exact size, shape and position (although sometimes it's helpful to think that it is). It does not orbit around the nucleus at a fixed distance like the earth orbits the sun (although sometimes it's helpful to think that it does)



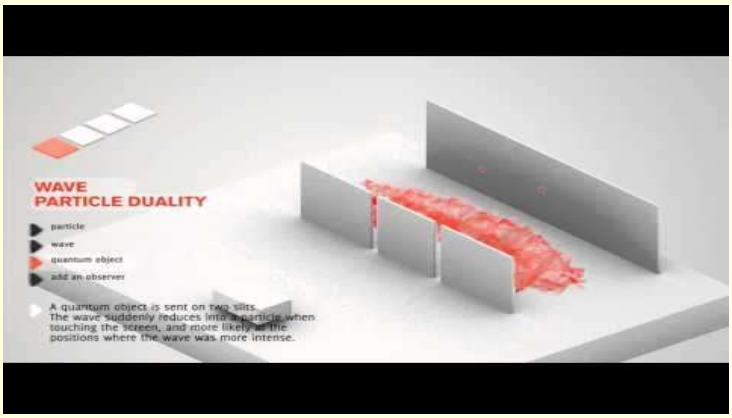
A fundamental property of quantum mechanics is that matter has wave-like properties (**wave-particle duality**).

Thus, instead of being just a discreet particle an electron can be described by a:

wavefunction

– a mathematical description of the distribution of electrons in terms of position and time.

Wave-Particle Duality



https://www.youtube.com/watch?v=Xmq_FJd1oUQ

An implication of describing the electron as a wave is the:

Heisenberg Uncertainty Principle

– we cannot know its *exact* position and momentum (energy) simultaneously.

As we know the exact energy of the electron it follows that we cannot know exactly where is it!



An implication of describing the electron as a wave is the:

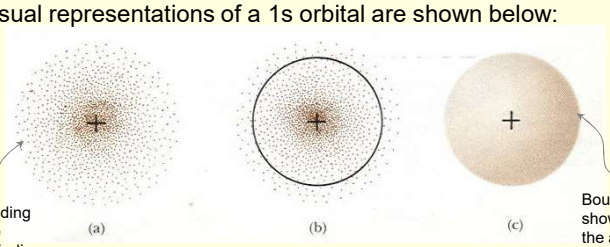
Heisenberg Uncertainty Principle

– we cannot know its exact position and momentum (energy) simultaneously.

As we know the exact energy of the electron it follows that we cannot know exactly where is it!

However we can determine where it is **most probably** located and that region of space is defined as an **orbital**.

Some visual representations of a 1s orbital are shown below:

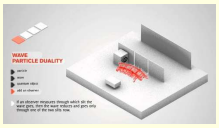


Density of shading represents the probability of finding the electron in that particular location

Boundary surface shows the outside of the area in which the electron is most likely to be located.

Exact location of electron never known – only where it's most probably located.

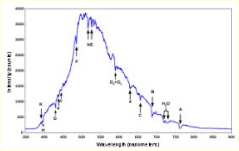
Quick Recap:



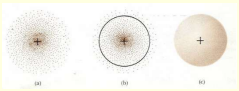
Electrons – have wave-particle duality



Heisenberg Uncertainty Principle – we cannot know an electron's position and momentum (energy) simultaneously.



Spectroscopy – we know the exact energy of electrons, therefore we cannot know exactly where they are.



Orbitals – describe where an electron of a specific energy is most probably located

We describe the orbital shapes and sizes using some labels called:
QUANTUM NUMBERS

1. The first orbital label is its '**shell'** or **principal quantum number (*n*)**.

Shells are of increasing size (or average distance from the nucleus).

Each electron in an atom is labelled with four quantum numbers. The first is the Principal quantum number: ***n*** (the '**shell**'):

***n* = 1, 2, 3, …**

The principal quantum number tells you about how big the orbital is, or how far (on average) an electron occupying it is from the nucleus (which is always at the centre of the orbital).

Defn: Integer value = whole number (e.g. 1, 2, 40, 103)

Within each shell there can be more than one **subshell.**

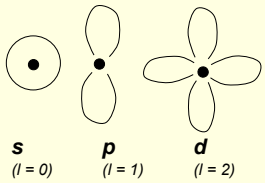
These are **labelled according to their shape: *s*, *p*, *d*, ..**

2. The second quantum number is the: **Angular Momentum Quantum Number (*l*)**

$l = 0, \ldots n-1$

This tells you about the **shape of the orbital:**
l = 0 is an "s orbital",
l = 1 is a "p orbital",
l = 2 is a "d orbital", etc.

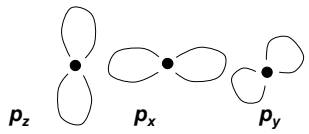
[***n*** = principal quantum number]



For *n* = 1 : only one subshell, **1*s***
For *n* = 2 : two subshells, **2*s*** and **2*p***
For *n* = 3 : three subshells, **3*s***, **3*p*** and **3*d***
Etc!

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 ***s***	Each ***s*** subshell contains: **one *s* – orbital**
 ***p*** $_z$ ***p*** $_x$ ***p*** $_y$	Each ***p*** subshell contains: **three *p* – orbitals**
See later…	Each ***d*** subshell contains: **five *d* - orbitals**

3. The third quantum number is the **magnetic quantum number (m_l)**:

$m_l = -l, \ldots +l$

e.g. if $l = 1$, $m_l = -1, 0, +1$

These are the p_x, p_y, and p_z orbitals, and tell you about the **orientation of the orbital**

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This table summarises the last few slides

n	*l*	Orbital notation	m_l	Orbital names
1	0	*1s*	0	*1s*
2	0	*2s*	0	*2s*
	1	*2p*	-1, 0, +1	$2p_x$, $2p_y$, $2p_z$
3	0	*3s*	0	*3s*
	1	*3p*	-1, 0, +1	$3p_x$, $3p_y$, $3p_z$
	2	*3d*	-2, -1, 0, +1, +2	$3d_{xy}$, $3d_{yz}$, $3d_{zx}$, $3d_{x^2-y^2}$, $3d_{z^2}$

etc!

Principal quantum number = distance from nucleus

n	*l*	Orbital notation	m_l	Orbital names
1	0	*1s*	0	*1s*
2	0	*2s*	0	*2s*
	1	*2p*	-1, 0, +1	$2p_x$, $2p_y$, $2p_z$
3	0	*3s*	0	*3s*
	1	*3p*	-1, 0, +1	$3p_x$, $3p_y$, $3p_z$
	2	*3d*	-2, -1, 0, +1, +2	$3d_{xy}$, $3d_{yz}$, $3d_{zx}$, $3d_{x^2-y^2}$, $3d_{z^2}$

etc!

Angular momentum quantum number = shape (type) of orbital

n	*l*	Orbital notation	m_l	Orbital names
1	0	*1s*	0	*1s*
2	0	*2s*	0	*2s*
	1	*2p*	-1, 0, +1	$2p_x$, $2p_y$, $2p_z$
3	0	*3s*	0	*3s*
	1	*3p*	-1, 0, +1	$3p_x$, $3p_y$, $3p_z$
	2	*3d*	-2, -1, 0, +1, +2	$3d_{xy}$, $3d_{yz}$, $3d_{zx}$, $3d_{x^2-y^2}$, $3d_{z^2}$

etc!

Magnetic quantum number = orientation of orbital

n	*l*	Orbital notation	m_l	Orbital names
1	0	*1s*	0	*1s*
2	0	*2s*	0	*2s*
	1	*2p*	-1, 0, +1	$2p_x$, $2p_y$, $2p_z$
3	0	*3s*	0	*3s*
	1	*3p*	-1, 0, +1	$3p_x$, $3p_y$, $3p_z$
	2	*3d*	-2, -1, 0, +1, +2	$3d_{xy}$, $3d_{yz}$, $3d_{zx}$, $3d_{x^2-y^2}$, $3d_{z^2}$

etc!

n	*l*	Orbital notation	m_l	Orbital names	
1	0	*1s*	0	*1s*	1 orbital
2	0	*2s*	0	*2s*	4 orbitals
	1	*2p*	-1, 0, +1	$2p_x$, $2p_y$, $2p_z$	
3	0	*3s*	0	*3s*	9 orbitals
	1	*3p*	-1, 0, +1	$3p_x$, $3p_y$, $3p_z$	
	2	*3d*	-2, -1, 0, +1, +2	$3d_{xy}$, $3d_{yz}$, $3d_{zx}$, $3d_{x2-y2}$, 3_{z2}	

etc!

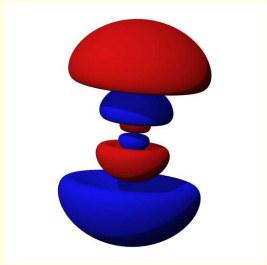
Q3. What are the quantum numbers for the three 4p orbitals (4px, 4py, 4pz - one is shown below)?

a. $n = 4$; $l = 0$; $m_l = -1, 0, +1$

b. $n = 4$; $l = 1$; $m_l = 3$

c. $n = 4$; $l = 1$; $m_l = 1$

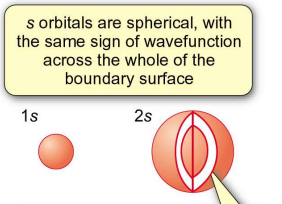
d. $n = 4$; $l = 1$; $m_l = -1, 0, +1$



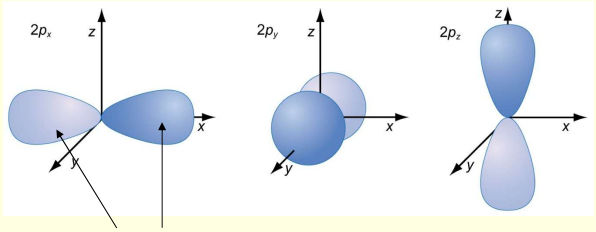
$\boldsymbol{n} = 1, 2, 3, \ldots$ | $\boldsymbol{l} = 0, \ldots n\text{-}1$ | $\boldsymbol{m_l} = -l, \ldots +l$

The shapes of orbitals are calculated from wavefunctions that describe the electron:

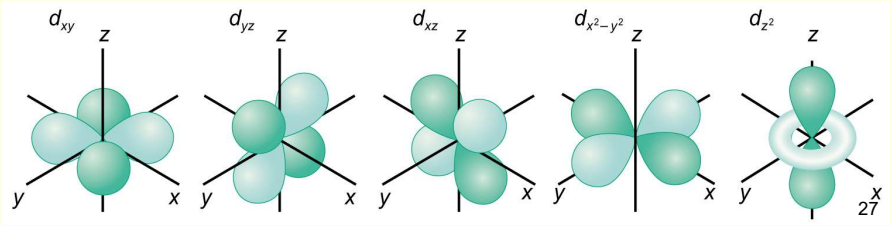
s orbitals are spherical, with the same sign of wavefunction across the whole of the boundary surface



The section taken out from a 2*s* orbital shows there is a spherical node. The wavefunction has a different sign either side of the node.



Different phases of wavefunction



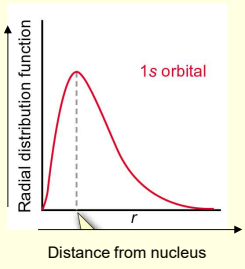
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So how do the wavefunctions explain the order of energies of the orbitals?

Energy is related to the distance of the electron from the nucleus. The further the orbital is from the nucleus, the higher the energy.

Wavefunctions can be used to calculate the **radial distribution function** of the orbital i.e. how the probability of finding the electron varies with distance from the nucleus.

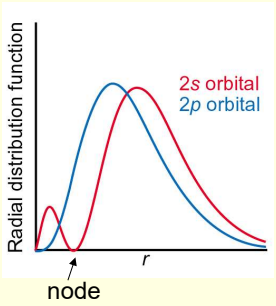
Probability of electron being found at that distance, r

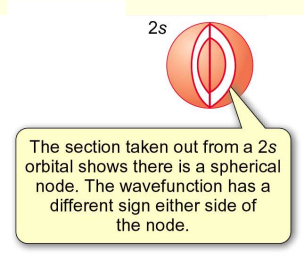


For the 1s orbital it can be seen that the electron has the highest probability of being found at the dotted line in the above figure. It also has a zero probability of being found exactly at the nucleus, but the chances increase rapidly with r, until the maximum. The probability of the electron being found a long, long way from the nucleus is very low – tending to zero.

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These distribution functions can be calculated for all of the orbitals and explain the relative energies of the 2s and 2p orbitals:

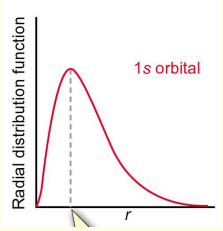


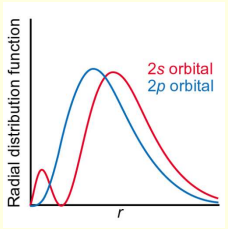


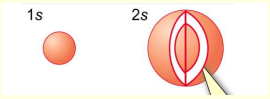
The energy of the orbital depends on the strength of the nuclear charge that the electron feels when it occupies that orbital.

29

29







- The $1s^2$(core electrons) shield the outer $2s^1$ electron from the nuclear charge slightly, BUT the 2s orbital radial distribution has a 'hump' or small maximum close to the nucleus
- There is some probability that the electron can 'penetrate' the core $1s^2$ orbitals and spend some time closer to the nucleus. When it does that, it experiences a stronger attraction to the nucleus than it would if it spent all of its time outside the 1s shell and hence its energy is lowered.
- If the electron were in the 2p orbital, the radial distribution function shows that there is very low probability of finding the electron close to the nucleus.
- This means that an electron in a 2p orbital does not penetrate the core electrons as much, and hence is shielded better from the nuclear charge. This means it has a higher energy.

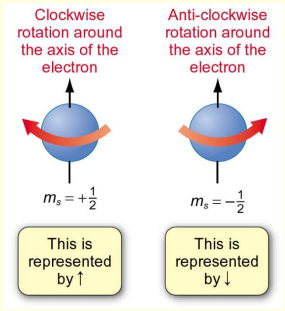
30

30

So we have explained the ordering of the orbitals but how do we explain why only two electrons can occupy each orbital?

4. Spin quantum number (m_s)

We need to consider that electrons have **spin**. This property of the electron is related to its angular momentum and its helpful to go back to considering the electron as a particle to visualise this:



m_s = +½ , -½

[This model is not really true as spin is a purely quantum mechanical property and a physical description is not possible.]

Which leads us to the **Pauli exclusion principle**:

The principle (Proposed by Wolfgang Pauli in 1925):

No two electrons in an atom can have the same four quantum numbers

The implication:



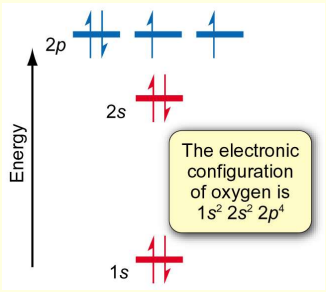
*A maximum of **two electrons can share each orbital**, with the same values of n, l, m_l but different spin (m_s = +1/2 and -1/2).*

Another thing we haven't considered is what happens when there are several orbitals of the same energy available. Which orbitals become occupied first?

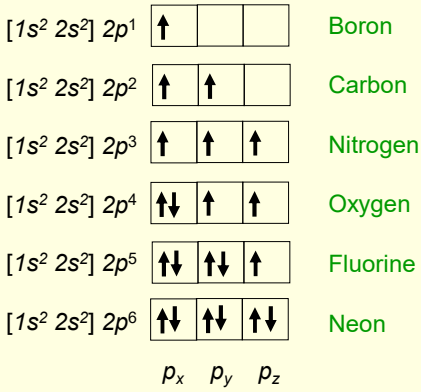
Hund's Rule *of maximum multiplicity*

*If two or more orbitals have the **same** energy, then electrons will spread out to occupy the maximum possible number of these, maximising the number of parallel spins.*

e.g.



Other first row atoms:



33

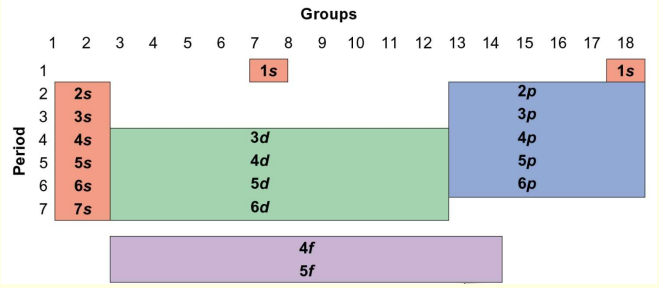
33

If the electrons occupy separate orbitals then they occupy different regions of space so there is less **electrostatic repulsion**. This is a lower energy arrangement.

The electrons have lower energy if their spins are parallel as a consequence of a quantum mechanical phenomenon called **spin correlation** – parallel spins will stay further away from each other and so repulsion is reduced.

The rules introduced above help to explain the electronic configuration of the different elements that were determined experimentally.

The elements are arranged in the periodic table in a reflection of their electronic configuration and in particular the identity of their **valence electrons**.



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The valence electrons of an element determine its **chemical reactivity** and the role it will play in biological function:

s block elements form stable cations e.g. Na^+ and Ca^{2+} as electrolytes in blood

p block elements can form multiple bonds e.g. C, O, N in sugars and proteins

d block elements available in range of oxidation states e.g. Cu in cytochrome c oxidase (could be harmful outside protective protein environment of enzyme)

Further reading

The material in this lecture is covered in Chemistry^3 – use the index to find the relevant sections.

In preparing this lecture I made use of 'Inorganic Chemistry' by D.F. Shriver and P.W Atkins (Chapter 1 of the 2nd Edition) but any general Inorganic or Physical Chemistry textbook will cover the same material. Find one in the library – an old edition is fine, this material hasn't changed over the past decade or so!

For more on Quantum Mechanics for those that are keen: 'Quanta: A Handbook of Concepts' by P W Atkins.

Please post any questions on Moodle – everyone is welcome to try to answer them!

Structure, Bonding, and Self-Assembly of Biological Structures

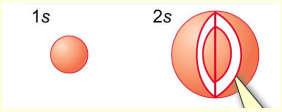


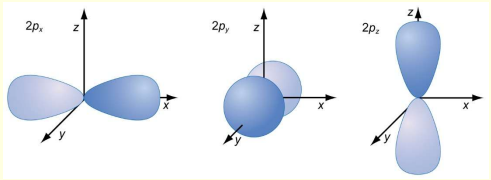
CHEM0010/1 – Part A

Dr. Anna Roffey
Academic Support Hours:
Mon 11:30-12:30
Thurs 2-3pm (CIB 124h)

Lecture 2

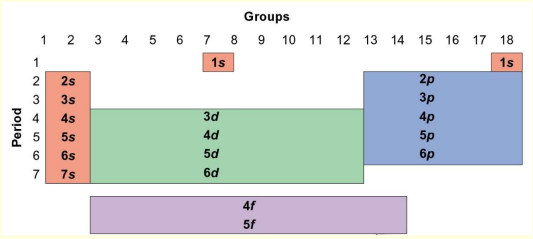
The shapes of orbitals are calculated from wavefunctions that describe the electron:





Electron configuration:

Helium Z = 2	$1s^2$
Carbon Z = 6	$1s^2\ 2s^2\ 2p^2$
Potassium Z = 19	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$ or [Ar] $4s^1$



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Learning Objectives: Atomic Structure and Covalent Bonding

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4. To introduce **orbital overlap** and **hybridisation** models for bonding (Valence Bond Theory). To draw single, double and triple bonds, and rationalise these using diagrams for orbital overlap.

5. To introduce basic **Molecular Orbital Theory**, as an alternative (and sometimes more accurate) representation of bonding in molecules, and apply this to diatomic molecules.

38

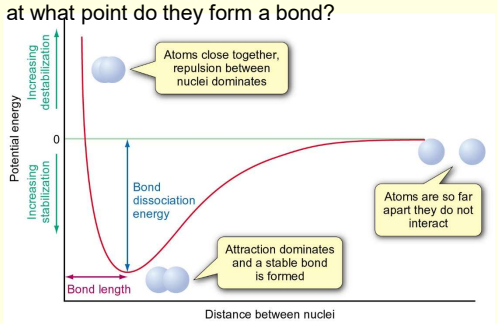
38

Covalent Bonding

So we have just learnt about the structure of atoms, but a question that has occupied chemists for years (and still occupies chemists today) is how best to explain the nature of a covalent bond.

What is a covalent bond?

If we imagine two hydrogen atoms separated in space and we envision moving them closer and closer together, at what point do they form a bond?



The potential energy of two hydrogen atoms varies with the distance between them. A pair of atoms are considered to have a chemical bond between them when they lie in the 'energy well', i.e. at lower energy than the individual atoms.

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Lewis structures – a first model for covalent bonding



[Contribution from the Chemical Laboratory of the University of California.]

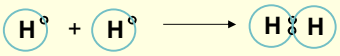
THE ATOM AND THE MOLECULE.

By Gilbert N. Lewis.

Received January 26, 1916.

The Lewis Model:

Covalent bonding occurs when valence electrons are shared between two atoms.



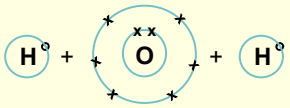
The maximum stability is achieved when each atom has a 'filled valence shell'. In this case each H has a full 1s orbital.

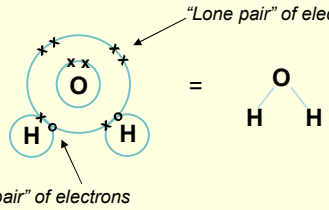
40

Lewis found he could account for a wide range of molecules using the **octet rule**:
each atom acquires shares in electrons until its valence shell achieves 8 electrons

H_2O

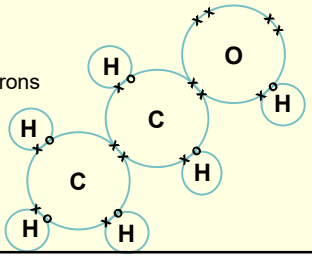
O $[1s^2]\ 2s^2\ 2p^4$ – 6 valence electrons
H $1s^1$ – 1 valence electron

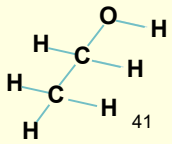




CH_3CH_2OH

C $[1s^2]\ 2s^2\ 2p^2$ – 4 valence electrons





"Core" electrons (1s) not shown

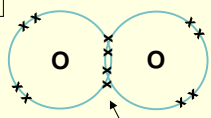
41

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Lewis structures – **double and triple bonds**

Explains why some bonds are stronger and shorter than other bonds

O_2



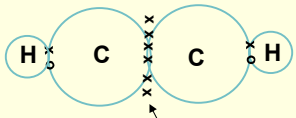
=



Double Bond – two shared pairs

* Later we will see that the O=O double bond is not so simple as this, and in fact is a rather special example ....

HCCH Ethyne / acetylene



=



Triple Bond – three shared pairs

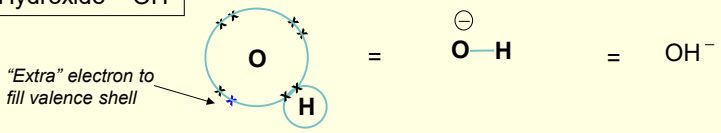
42

42

Lewis structures – **charged molecules (ions)**

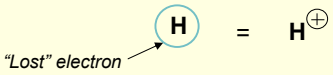
Molecules can accept or lose electrons to give a complete valence shell. The result is a negative or positive ion.

Hydroxide – OH^-



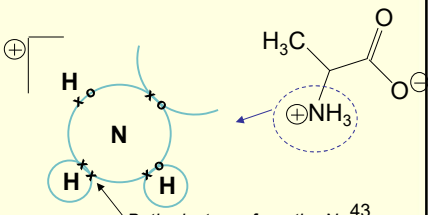
Hydrogen ion – H^+

No electrons left! H^+ is just a proton!



Coordinate bonds, e.g. ammonium in Alanine

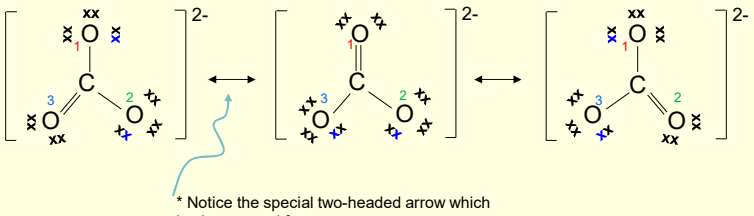
$R\text{-}NH_2 + H^+ \longrightarrow R\text{-}NH_3^+$



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Lewis structures – **'resonance hybrids'**

Sometimes it is possible to draw more than one satisfactory Lewis structure for a particular molecule. The true structure can then be considered to be an average of the Lewis structures: a 'resonance hybrid' with contributions from different 'resonance forms'.



Carbonate – CO_3^{2-} *24 valence electrons*

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Hypervalency

Although period 2 elements obey the octet rule very well, Period 3 elements begin to show some deviations.

e.g. PCl_5 can only be explained if P has 10 electrons in its outer shell

In SF_6 S must have 12 electrons in its outer shell

Compounds which require more than an octet of electrons in order to draw a Lewis structure are called **hypervalent compounds**.

Traditionally this is explained by the availability of low-lying d-orbitals to accommodate the extra electrons.

We will see in the next lecture how we can use another model to explain hypervalency without involving the d-orbitals.

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Prediction of Molecular Shape using VSEPR theory (**v**alence **s**hell **e**lectron **p**air **r**epulsion)

The Lewis model provides a good starting point for explaining which compounds are formed. In the following decades the model was used to explain the **shapes** and **symmetry** of molecules.

BAKERIAN LECTURE

Stereochemical types and valency groups

BY N. V. SIDGWICK, F.R.S. AND H. M. POWELL

(*Delivered* 27 *June* 1940—*Received* 27 *June* 1940)

Sidgwick and Powell, *Proc. Roy. Soc.*, 1940, ***A***, **176**, 153.

INORGANIC STEREOCHEMISTRY

By R. J. GILLESPIE, D.Sc., and R. S. NYHOLM, M.Sc., D.Sc.

(UNIVERSITY COLLEGE, LONDON, W.C.1)

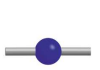
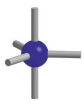
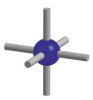
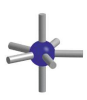
(Quarterly reviews of the Chemical Society, 1957)

Molecular Shape: using VSEPR theory

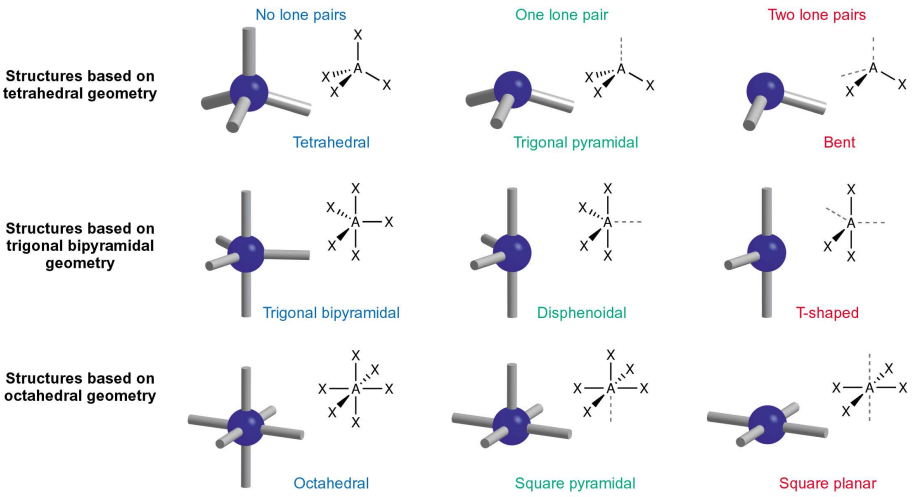
Basic assumptions:

The electrons in bonds and lone-pairs around an atom can be considered as 'charge clouds' which repel each other.

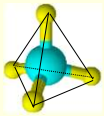
The lowest energy arrangement is when they are as far apart as possible, and this determines the equilibrium molecular shape.

Number of electron pairs around central atom	2	3	4	5	6	7
	Linear	Trigonal planar	Tetrahedral	Trigonal bipyramidal	Octahedral	Pentagonal bipyramidal
Shape						
Bond angles	X—A—X 180°	120°	109.5°	90°, 120°	90°	72°, 90°
Example	BeF_2	BF_3	CF_4	PF_5	SF_6	IF_7

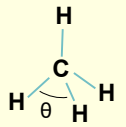
Lone pairs must also be taken into account when predicting the shape of the molecule:

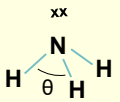


Another assumption is that <u>lone pairs repel more than bonding pairs</u>

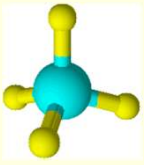


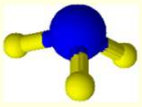
Examples which are all based on TETRAHEDRAL geometry (total bonds + lone pairs = 4):

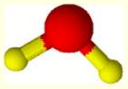












θ = 109.5°

'tetrahedral'

θ = 107.3°

'trigonal pyramidal'

θ = 104.5°

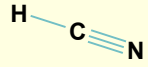
'bent'

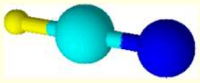
Bond angle gets smaller due to lone pair repulsion

According to the VSEPR model, a multiple bond is treated as through it was a single electron pair:

Examples of LINEAR geometry (total bonds + lone pairs = 2):

HCN

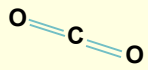


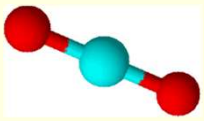


ɵ = 180°

'linear'

CO_2

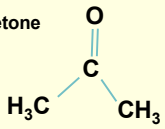


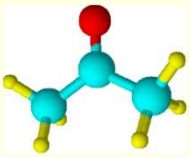


ɵ = 180°

'linear'

Acetone

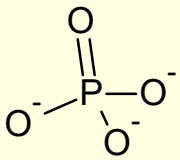


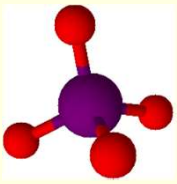


ɵ = 120°

'trigonal planar'

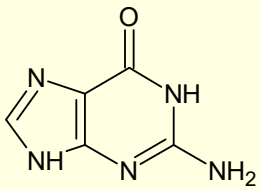
This means that we don't have to worry about any resonance structures:

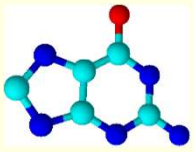




tetrahedral

VSEPR also explains why molecules like nucleobases (e.g. guanine) are planar – important in biological function and how they stack in nucleic acids like DNA.

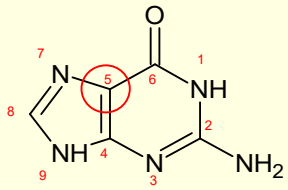




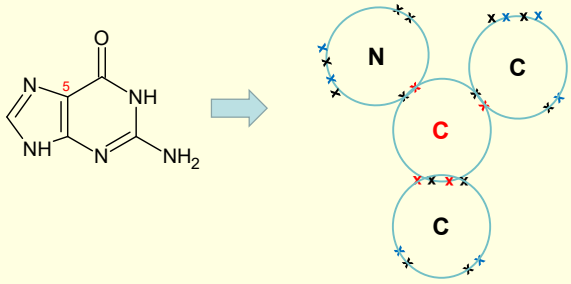
What is the geometry around the carbon atom at position 5 in the nucleobase molecule guanine (below)?

Hint:

1. Draw the Lewis structure for that carbon atom and its immediate bonds.
2. Count the number of bonds (treat multiple bonds as single).
3. Are there lone pairs? If so, add them to the total.
4. Go back to slides 48 and 49 and find the corresponding geometry for that combination of bonds and lone pairs.

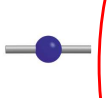
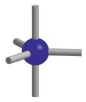


1. Draw the Lewis structure for that carbon atom and its immediate bonds.

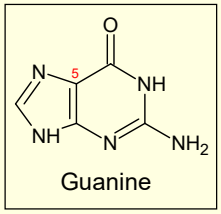


2. Count the number of bonds (treat multiple bonds as single). **= 3**
3. Are there are lone pairs? If so add them to the total. **= 0**
4. Go back to slides 48 and 49 and find the corresponding geometry for that combination of bonds and lone pairs.

Molecular Shape: using VSEPR theory

Number of electron pairs around central atom	2	3	4	5	6	7
	Linear	Trigonal planar	Tetrahedral	Trigonal bipyramidal	Octahedral	Pentagonal bipyramidal
Shape						
Bond angles						
Example	BeF_2	BF_3	CF_4	PF_5	SF_6	IF_7

Guanine is planar – important in biological function and how nucleotides stack in DNA.



Guanine

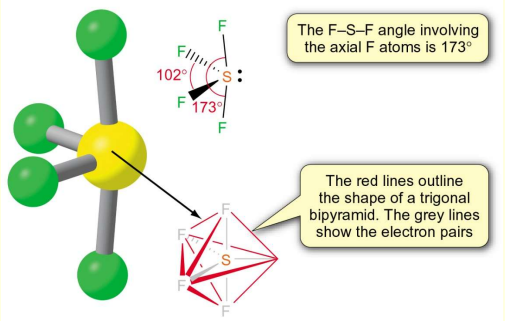
55

55

Another assumption is that if a lone pair has a choice between an equatorial position and an axial position, it will occupy the equatorial site.

This is because in the equatorial position it is repelled less by the two axial bonding pairs than it would be by the three equatorial bonding pairs if it was in the axial position.

For example, SF_4 molecule:



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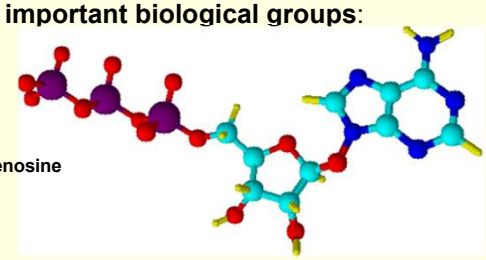
56

Bonding and geometry of some important biological groups:

ATP (adenosine triphosphate) exemplifies many of these geometric bonding structures

ATP is made up from three sections:
A purine base – **Adenine**
A pentose sugar – **Ribose**
A phosphate section - **Triphosphate**

} **= Adenosine**



Importance of molecular shape in biology

- Active sites of enzymes specific for substrate of particular size and shape = greater selectivity
- Cell receptors / signalling pathways / responses to hormones depend on molecular recognition
- Antibody / antigen interactions
- If the wrong molecule binds to an enzyme active site this can lead to problems, e.g. CN^-, N_3^-, CO all inhibitors of cytochrome c oxidise active site, as compete with O_2 for binding.

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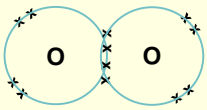
57

Limitations of the Lewis model of bonding

The simple Lewis model is a very powerful tool for explaining many trends in bond lengths, reactivity and molecular shape. Its simplicity means that it is a tool that can used daily by chemists for simple predictions and to aid our understanding.

However, the model has some limitations. It cannot explain **hypervalency** very satisfactorily, as discussed previously.

It also does not explain some other observations, for example that the oxygen molecule is **paramagnetic** (it has unpaired electrons), which is not predicted by the Lewis model.



Next, we will look at more complex models for bonding that use the overlap of atomic orbitals to explain the formation of covalent bonds.

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Structure, Bonding, and Self-Assembly of Biological Structures



CHEM0010/1 – Part A

Dr. Anna Roffey

Academic Support Hours:
Mon 11:30-12:30
Thurs 2-3 pm (CIB 124h)

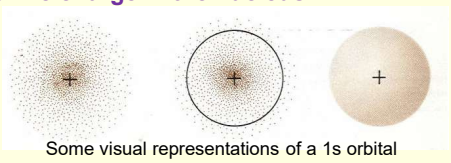
Lecture 3

Recap of Lecture 1

Energy is related to the distance of the electron (e^-) from the nucleus – strength of positive charge the e^- feels.

The further the orbital is from the nucleus, the higher the energy of an e^- in that orbital – **the less stabilised it is by the +ve charge in the nucleus**.

Remember, an orbital is a probability distribution for the location of an e^- in space, a certain distance from the nucleus.

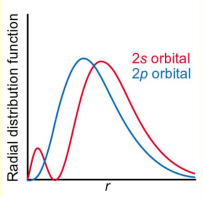


Some visual representations of a 1s orbital

The energy of the orbital depends on the strength of the nuclear charge that the e^- feels when it occupies that orbital.

Electrons can be shielded from the +ve charge by other e^-s in lower energy orbitals.





Learning Objectives: Atomic Structure and Covalent Bonding

1. ATOMIC STRUCTURE: To recall the **electron configuration of atoms**, and how patterns of electron configuration are represented in the periodic table.
2. To understand and predict the COVALENT BONDING in simple molecules made up of C, O, H, N, S, P using **electron-sharing** arguments (Lewis structures).
3. To understand the geometry of covalently bonded molecules based on **valence shell electron pair repulsion** (VSEPR).
4. To introduce **orbital overlap** and **hybridisation** models for bonding (Valence Bond Theory). To draw single, double and triple bonds, and rationalise these using diagrams for orbital overlap.
5. To introduce basic **Molecular Orbital Theory**, as an alternative (and sometimes more accurate) representation of bonding in molecules, and apply this to diatomic molecules.

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Further models for covalent bonding

Lewis structures, as discussed previously, are often useful for predicting or rationalising the stoichiometry and bonding of molecules.

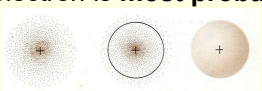
However Lewis's method is purely formal: it cannot tell us anything about what the sharing of electrons **means** or how it **arises**.

This is where Valence Bond Theory, Hybridisation, and Molecular Orbital Theory come in.

As stated in an earlier lecture:

Instead of being just a discreet particle, an electron can be described by a **wavefunction** – a mathematic description of the location of the electron in space.

The region in space where the electron is **most probably** located is defined as an **orbital**.



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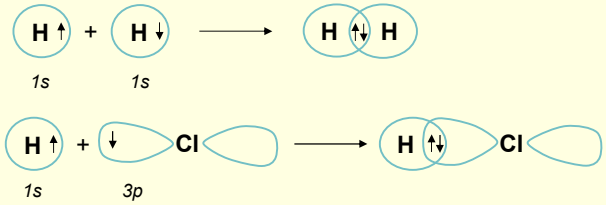
62

Valence-Bond Theory

Valence bond theory describes how half-filled atomic orbitals on two atoms overlap to create a bond containing paired electrons.

A bonding orbital is created between the two atoms.

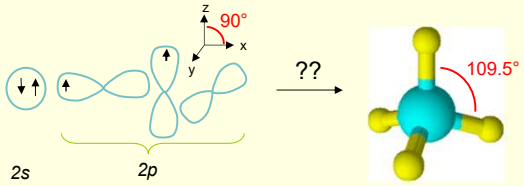
For example:

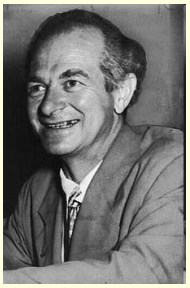


It is assumed that the **bonding electrons** are most likely located in the region **between the atoms**, giving rise to a bond.

But…..how do *s*, *p*, and *d* atomic orbitals contribute to covalent bonding? In particular, how do electrons from *s* and *p* atomic orbitals form equivalent covalent bonds, e.g. in CH_4 ?

Carbon: $1s^2\ 2s^2\ 2p^2$





Linus Pauling (1901 – 1994)
First described concepts of 'hybridisation' and 'electronegativity'.
Nobel prize in Chemistry 1954 & Nobel Peace Prize 1962

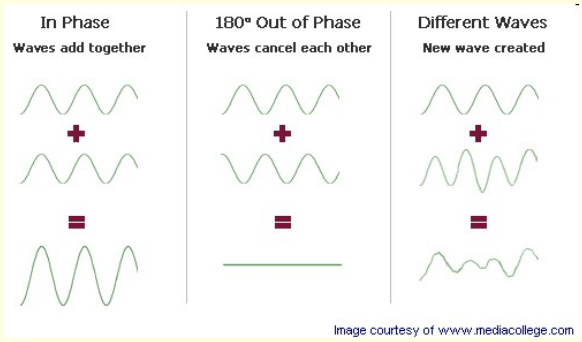
Linus Pauling proposed an answer to this problem, in terms of combinations of the different atomic orbitals to give the same number of equivalent hybrid orbitals.

Hybridisation

If we consider the carbon atom in methane, there are 4 valence atomic orbitals: s, p_x, p_y, p_z

Each of these orbitals can be described as a **wavefunction**, a mathematic description of the location of the electron in space.

As with any waves they can be combined both constructively and destructively:



So in methane, the C atomic orbitals combine in four different combinations (which you don't need to remember for the exam!):

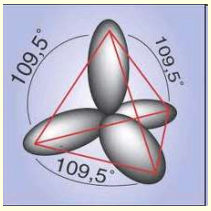
$H_1 = s + p_x + p_y + p_z$

$H_2 = s - p_x - p_y + p_z$

$H_3 = s - p_x + p_y - p_z$

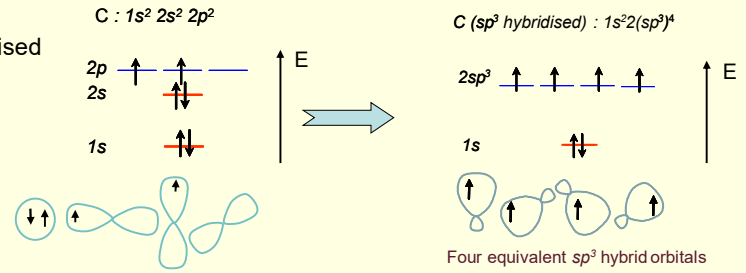
$H_4 = s + p_x - p_y - p_z$

These equivalent orbitals are called **sp^3-hybrid orbitals** and are directed towards the corner of a regular tetrahedron

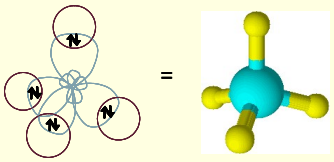


The four sp^3 orbitals are the same energy and each contains a single electron:

Unhybridised carbon



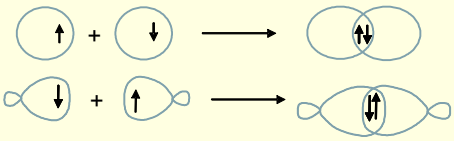
To form methane, each of the sp^3 orbitals overlaps with a 1s orbital of H, to form a bond:



These bonds are called '**sigma**' or **'σ' bonds**.

They lie along the line between the atoms.

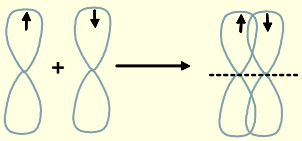
They are 'symmetric to rotation about the bond'.



two s-orbitals forming a σ-bond

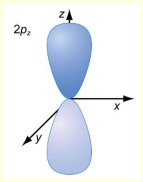
two hybrid orbitals forming a σ-bond

Sometimes orbitals form bonds by overlapping 'side-ways on'



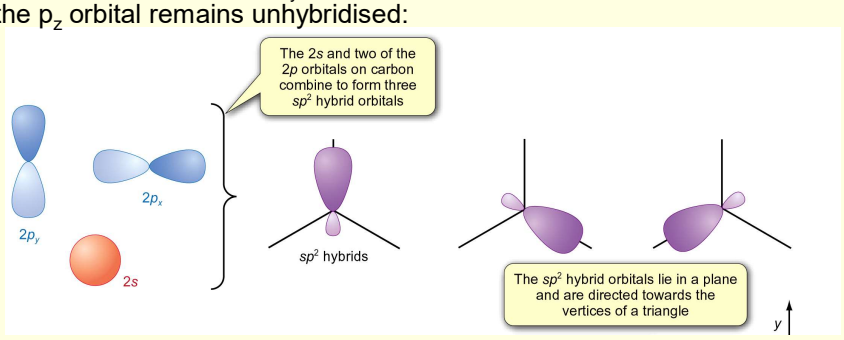
These bonds are called 'pi' or 'π-bonds'. They have two lobes, above and below the line between the atoms.

They have the same symmetry as p-orbitals. This means they are not symmetric to rotation about the bond.

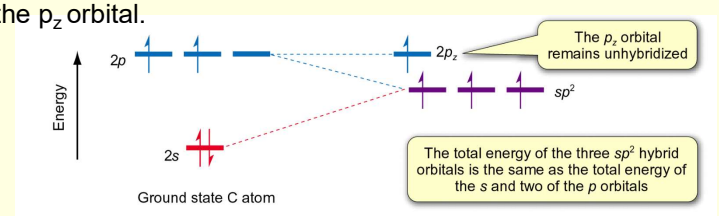


Molecules such as ethene have both σ and π bonding. The reason for this is that the carbon in ethene is **sp^2 hybridised**.

In ethene, the s, p_x and p_y orbitals combine to form hybrid orbitals, while the p_z orbital remains unhybridised:



The valance electrons are distributed between the three new sp^2 orbitals and the p_z orbital.

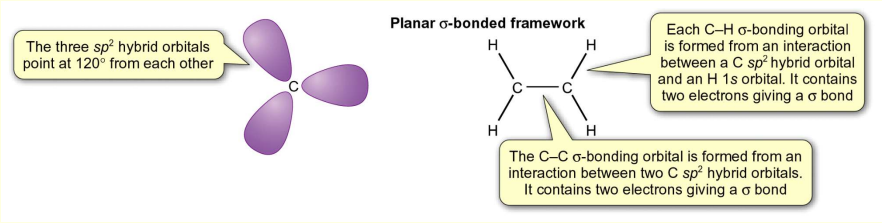


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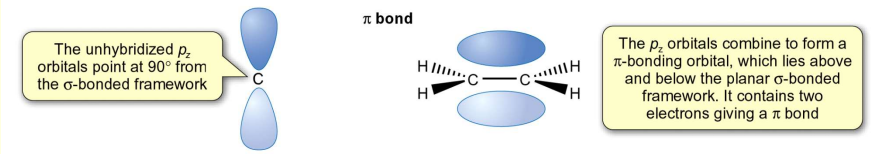
69

The two carbon atoms in ethene can form bonds in two ways:

1. An sp^2 orbital from each can overlap to form a σ-bond



2. The unhybridised p_z orbital on each C can overlap 'sideways on' to give a π orbital.



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Strengths and weaknesses of the valence bond model

Like the Lewis bond model, the valence bond model is helpful in explaining some observations about molecular shape and reactivity. It gives more insight than the Lewis model into 'what is a bond?'

These models are known as 'localised models' as they focus on the nature and properties of one particular bond in a molecule. They are very useful for discussion of bond length, bond strength, bond force constant (see Section B) and some aspects of reactivity, such as acid – base interactions and bond polarity.

However sometimes these models can't explain some observations (like the paramagnetism of O_2, as mentioned previously). Sometimes the properties of a molecule cannot be explained by considering the bonding of a single bond.

Global properties of some molecules such as electronic spectra (see Section B) and reduction (electrode) potentials (see Section D) need us to consider the electron distribution over the whole molecule. This is a delocalised electron description and is called **Molecular Orbital Theory** and we'll consider this now.

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Molecular Orbital Theory

Let's start with the most simple molecule…

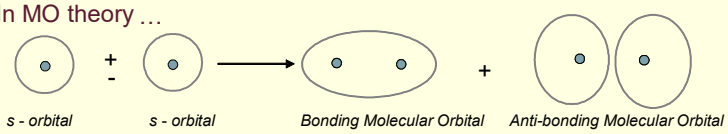
$$H + H \longrightarrow H_2$$

The valence orbital for each H atom is the 1s orbital, which as discussed previously can be described by a wavefunction.

Combinations of waves can either be constructive or destructive.

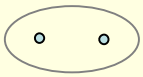
We use this argument when combining the s orbitals to from a bond. If they combine constructively then electron density between the atoms is increased. This is called a **bonding molecular orbital**. If the s orbitals combine destructively then electron density between the atoms is decreased and this is called an **anti-bonding molecular orbital**.

In MO theory …



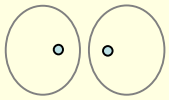
A Molecular Orbital (MO) describes the region in space where an electron is likely to be found.

Bonding MO



High probability of electron being found between atoms = MO contributes to bonding

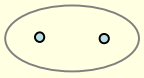
Anti-Bonding MO

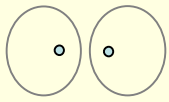


Although this looks like two orbitals – its just one with a node in the middle

Zero probability of electron being found between atoms = MO negates from bonding

The MOs are labelled according to their symmetry:

σ 

σ* 

Antibonding orbitals are always denoted with a *

What is the relative energy of the MOs?

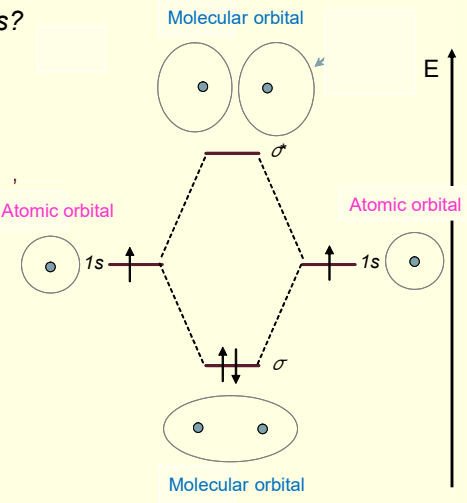
When the two atomic orbitals combine to form MOs, the bonding MO is lower in energy than the original orbitals and the anti-bonding MOs are higher in energy.

Where do the electrons go?

For H_2 there are two valence electrons, one from each H atom.

As for electronic configurations in atoms, we simply fill the lowest energy MO first.

If two electrons occupy the same MO they must have the opposite spin.



Molecular Orbital Diagram

Some MO Rules:

- There are always the same total **number of molecular orbitals** as contributing atomic orbitals

  2 Atomic orbitals = 2 Molecular orbitals (1 bonding, 1 antibonding)

- Only orbitals of the **same symmetry** will combine

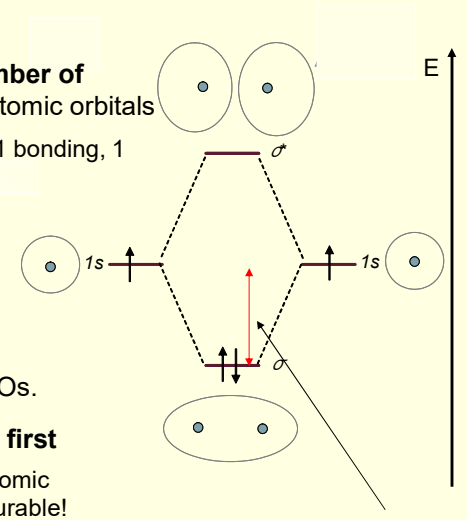
  In this case its s orbital + s orbital so OK

- Atomic orbitals need to be **close in energy** in order to overlap to form MOs.

- The electrons **fill lower energy MOs first**

  Here they are lower in energy than the atomic orbitals. So, the bonding of H + H is favourable!

- More **electronegative** elements how lower E AOs



Stability gained through forming the bond

What about He_2?

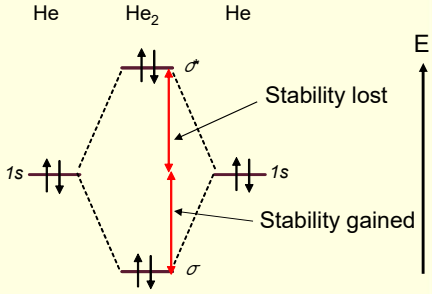
Same orbitals involved as H_2

But now 4 electrons

2 occupy bonding MO, 2 occupy antibonding MO

No energy gain from bonding

He exists as atoms rather than molecules



We can explain the stability of molecules by considering their **Bond Order**:

$$\text{Bond Order} = \frac{(\text{number of bonding electrons}) - (\text{number of anti-bonding electrons})}{2}$$

So for He_2: Bond Order = (2 – 2) / 2 = 0

A bond order of zero means there is no bond, so the He_2 molecule does not exist.

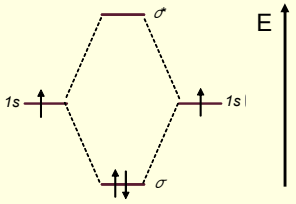
This tells us that electrons in an antibonding orbital cancel out electrons in a bonding orbital.

77

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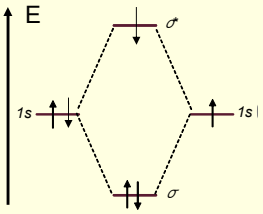
Bond order of H_2 = (2 – 0) / 2 = 1

A bond order of 1 is equivalent to a single bond using the Lewis model, and is consistent with observations.



What about the He_2^+ molecule? Does it exist?

He_2^+ is He_2 with one less electron.



Bond order of He_2^+ = (2 – 1) / 2 = ½

A bond order of ½ is something that is not encountered in the Lewis 'electron pair' model, but something that MO theory can predict with ease.

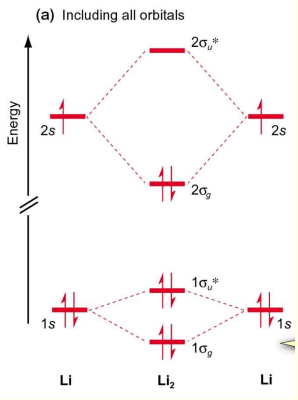
What this means is that a bond of about half the strength of a single bond can form between the He atoms of He_2^+

This means that this molecule can exist, but will not be very stable. Its existence is likely to be transient.

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Molecular orbitals for Li_2 and Be_2



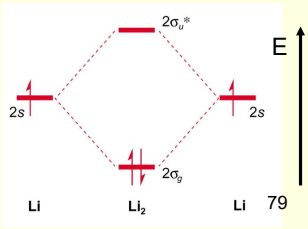
Atomic orbitals need to be **close in energy** in order to overlap to form MOs.

The 1s and the 2s orbitals therefore don't overlap with each other.

The 1s orbitals overlap to give 1σ and 1σ* MOs and the 2s orbitals overlap to give 2σ and 2σ* MOs.

The degree of overlap is reflected in the separation in energy of the bonding and antibonding MOs. The better the atomic orbital overlap, the lower the bonding MO is in energy (and the higher in energy the anti-bonding MO is).

This makes sense, as the better the overlap, the stronger the bond.



Bond order for Li_2 = (4 – 2) / 2 = 1 i.e. a single bond

MO diagrams only usually take into account the valence orbitals, so we can leave out the 1σ and 1σ* orbitals as they don't contribute to the bonding

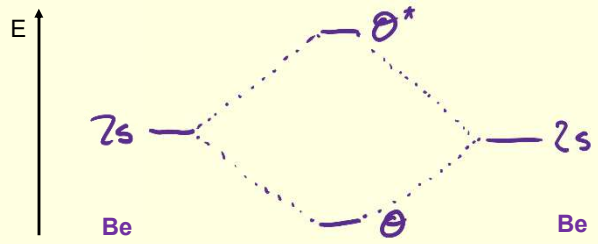
Q2. What is the bond order of Be_2 and does this molecule exist?



Hint:

1. Draw the Molecular Orbital diagram for Be_2.
2. Fill the atomic orbitals with the valence electrons.
3. Fill the molecular orbitals with the valence electrons.
4. Calculate the bond order.

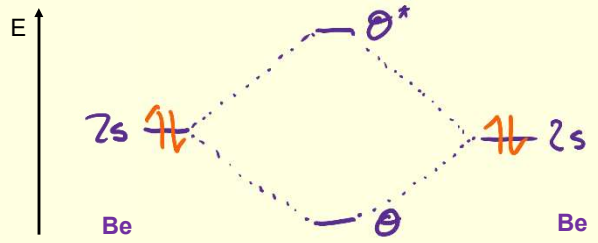
1. Draw the Molecular Orbital diagram for Be_2. (*Be: $1s^2$ $2s^2$*)



81

81

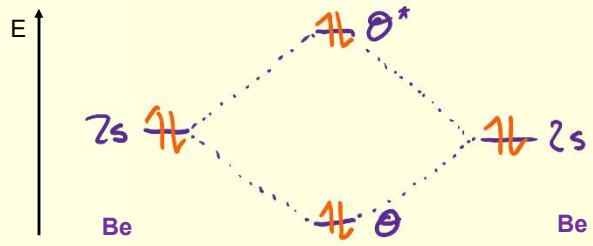
1. Draw the Molecular Orbital diagram for Be_2. (*Be: $1s^2$ $2s^2$*)
2. Fill the atomic orbitals with the valence electrons.



82

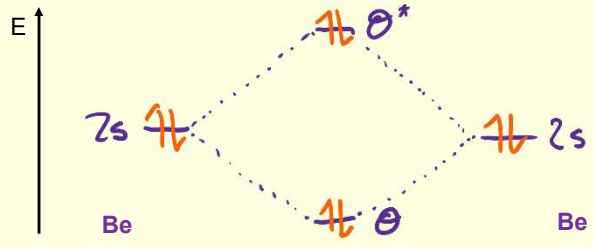
82

1. Draw the Molecular Orbital diagram for Be_2. (*Be: $1s^2$ $2s^2$*)
2. Fill the atomic orbitals with the valence electrons.
3. Fill the molecular orbitals with the valence electrons.



1. Draw the Molecular Orbital diagram for Be_2.
2. Fill the atomic orbitals with the valence electrons.
3. Fill the molecular orbitals with the valence electrons.
4. Calculate the bond order.

No, Be_2 does not exist!

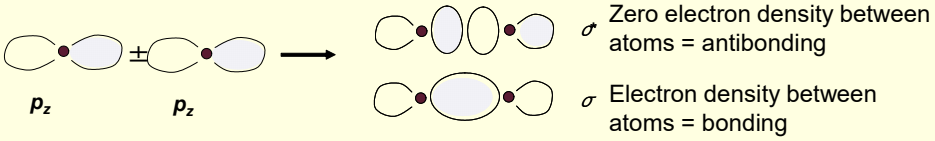


Bond order = $\frac{\text{bonding e}^- - \text{antibonding e}^-}{2} = \frac{0}{2} = 0$

How are MOs formed from p orbitals?

As discussed previously, p orbitals can form both σ and π bonds depending on how they overlap.

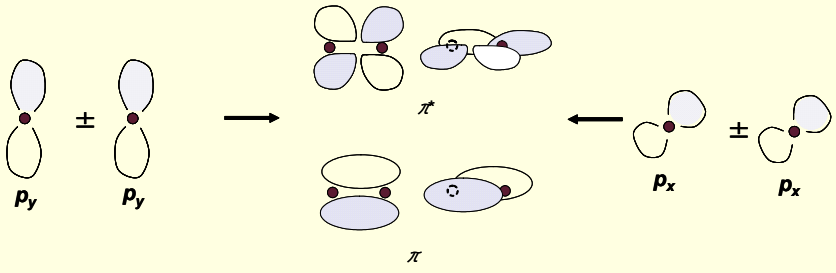
σ bonds are formed when they overlap head on:



Once again MOs can be formed through constructive interference of the wavefunctions (bonding MO) and through destructive interference (antibonding MO)

As the resulting MOs are symmetrical with respect to rotation about the bond they are labelled sigma.

If the p_z orbitals overlap head-on, the p_x and p_y orbitals can only overlap side-on and hence they form π MOs:

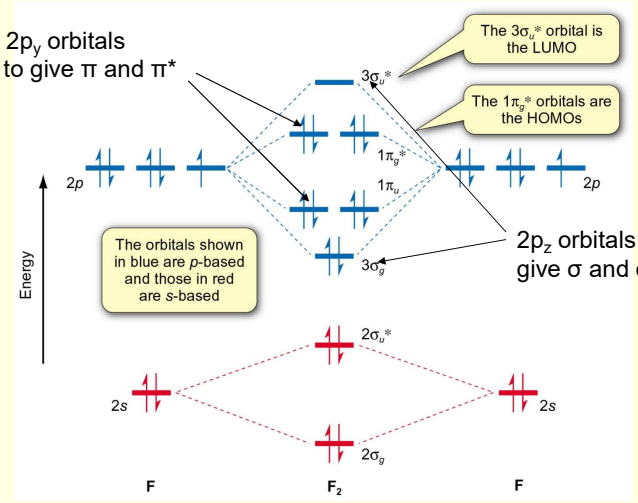


[When forming MOs the p_z by convention undergoes σ bonding and the p_x and p_y undergo π bonding. The observant among you may have noticed that in contrast in the theory of hybridisation the p_z partakes in π bonding and the other two (in the form of hybridised sp^2 orbitals) undergo σ bonding. I agree this is confusing but its just a convention thing!]

MOs for first row diatomic molecules

We're going to start by considering F_2 which has 14 valance electrons to consider

$2p_x$ and $2p_y$ orbitals overlap to give π and π* MOs



Lowest unoccupied MO

Highest occupied MO

$2p_z$ orbitals overlap to give σ and σ* MOs

This diagram is valid for O_2, F_2 and Ne_2 (with the relevant number of electrons of course)

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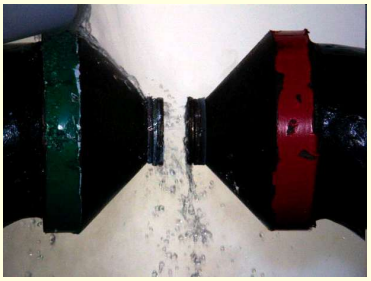
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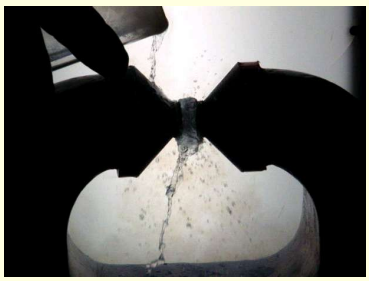
This highlights a disadvantage to MO theory. It is very difficult to predict the order in energy of orbitals except for the simplest of molecules. It is therefore not as useful a tool as Lewis Theory in some circumstances. Higher level computation and comparison to experimental data (e.g. electronic spectroscopy) is required.

However it is very powerful at revealing details about the electronic structure of molecules that Lewis theory cannot explain. For example:

Liquid Nitrogen is not attracted to a magnet

Liquid Oxygen is attracted to a magnet





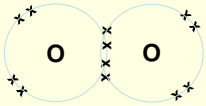
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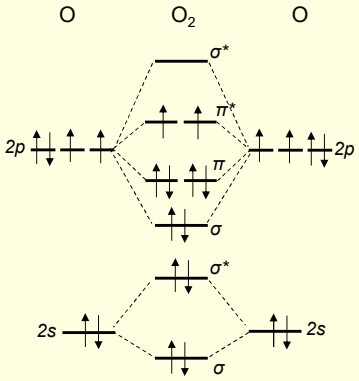
88

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Liquid oxygen is found to be paramagnetic – it therefore must possess **unpaired electrons**.

This is not predicted by Lewis theory:





However the MO diagram for O_2 reveals that the 1π* orbitals each contain an unpaired electron.

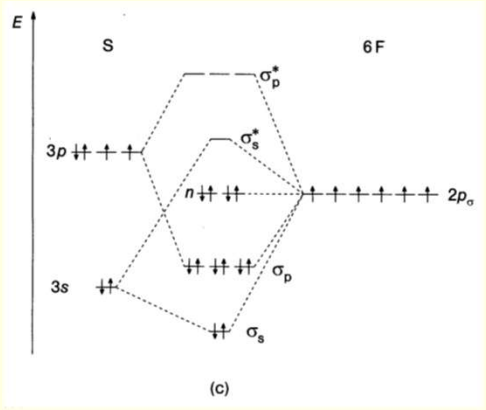
[Remember Hund's rule of maximum multiplicity]

In contrast the MO diagram from N_2 reveals that all of its electrons are paired.

Liquid N_2 is therefore not attracted to a magnet.

A last word on **hypervalency**…<u>that you do not need to know for exams!</u>

As mentioned previously, Lewis Theory does not deal very satisfactorily with hypervalency, e.g. in SF_6. It was assumed that low lying d-orbital contribute to bonding in that model. In fact, it is found that d-orbitals don't contribute at all. But the bonding can be explained more effectively by using MO Theory, which provides enough MOs to house all of the electrons.



Want more?

The material in this lecture is covered in Chemistry[3] – use the index to find the relevant sections

In preparing this lecture I made use of 'Inorganic Chemistry' by D.F. Shriver and P.W Atkins (Chapters 2 and 3 of the 2nd Edition) but any general Inorganic or Physical Chemistry textbook will cover the same material. Find one in the library – an old edition is fine, this material hasn't changed over the past decade or so!

Please post any questions on moodle – everyone is welcome to try to answer them!

Structure, Bonding, and Self-Assembly of Biological Structures

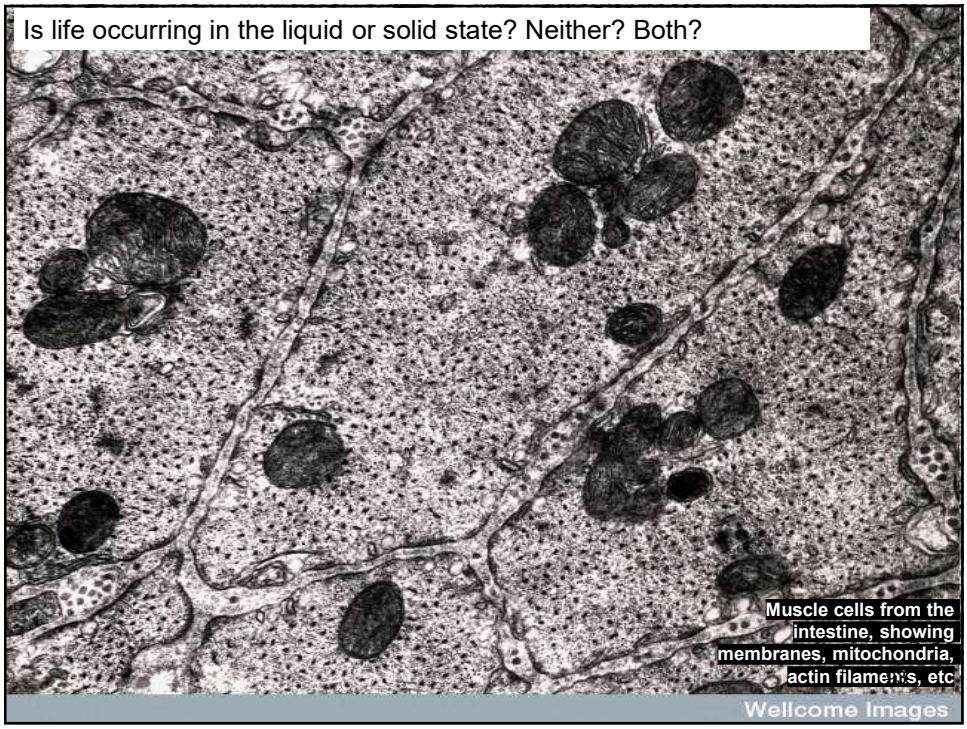
CHEM0010/1 – Part A

Dr. Anna Roffey
Academic Support Hours:
Mon 11:30-12:30
Thurs 2-3 pm (CIB 124h)

Lecture 4



Is life occurring in the liquid or solid state? Neither? Both?



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CHEM0010/11 Chemistry for Biologists
Section A: Structure and Bonding

Learning Objectives: Non-covalent molecular interactions

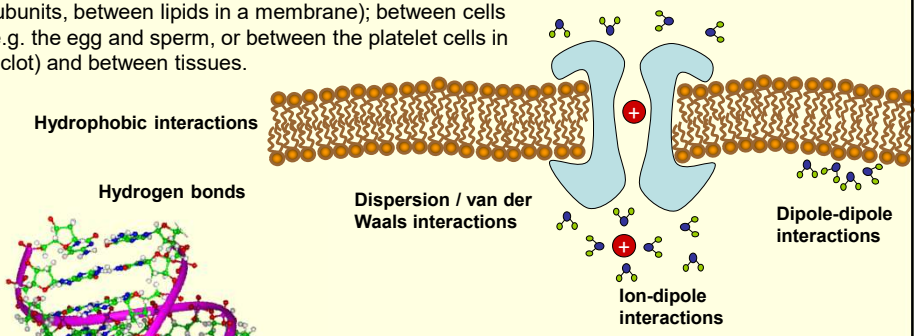
1. To have a broad introduction to the role of non-covalent interactions between molecules and their essential role in determining the properties of condensed-phase materials.
2. To recall the concept of electronegativity and bond polarity, then further to understand how molecular liquids have differing molecular **polarity**. The relationship between solubility and solvent polarity and polarisability.
3. To focus on the interesting properties of water as nature's solvent, in particular the **dissociation** and **hydration** of ionic species in water.
4. To understand (qualitatively) the **hydrogen bonding** within bulk water, between water and solute molecules, and within and between other molecules.
5. To understand the origin of **hydrophobic interactions** which cause the aggregation of non-polar moieties in water.
6. To understand the basic thermodynamic explanations for **self-assembly**, building from the non-covalent interactions studied previously
7. To encounter a range of **examples of biological self-assembly**.

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What are non-covalent interactions? (And why do we care about them?)

Non-covalent interactions explain much of the fine control within large molecules (e.g. protein folding, DNA helical structure); between molecules (between protein subunits, between lipids in a membrane); between cells (e.g. the egg and sperm, or between the platelet cells in a clot) and between tissues.

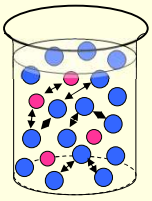


Where do these come from; how and when do they occur? This is the topic of the next lectures.

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The difference between covalent and non-covalent interactions





Covalent bonds:

- Link atoms within a molecule
- Strong bonds (~100s kJ/mol or ~100k_BT)
- Short ranged
- Highly directional / geometrically well defined
- Generally 'long lasting'

k_B = Boltzmann constant

= 1.38×10^{-23} J K^{-1}

See section C where this should become clear

Non-Covalent Interactions:

- Between neighbouring molecules (*intermolecular*), or non-bonded parts of the same molecule (*intramolecular*)
- Weak "interactions" (~k_BT)
- Longer range
- Less directional / geometrically well defined. *(depending on the type of n-c interaction)*
- Generally short lived or fluctuating *(though not always…)*

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Different types of non-covalent interactions

Table 1.10 Types of non-covalent interactions. For comparison, the energies of covalent bonds lie in the range 150 kJ mol^{-1} to 1000 kJ mol^{-1}.

Interaction	Acts between	For example	Typical energy / kJ mol^{-1}
London dispersion	All types of molecules	H_2, CH_4, Cl_2, HCl	~ 5
Dipole–dipole	Polar molecules	HCl	2 } in gas phase
Dipole–induced dipole	A polar molecule and a molecule that may or may not be polar	Between HCl and Cl_2 or between HCl and CH_4	< 2
Ion–dipole	An ion and a polar molecule	A metal ion dissolved in water	15
Hydrogen bond	A molecule containing the electronegative atoms O, N, or F bound to H and a molecule containing O, N, or F. The H atom forms a link between the two electronegative atoms	H_2O, NH_3, HF	10–40*
Ion–ion	Ionic groups	Between $-NH_3^+$ and $-CO_2^-$ groups in proteins	250

* The hydrogen bond in the ion, $[HF_2]^-$, is particularly strong, 165 kJ mol^{-1}; see Section 25.3 (p.1147).

Learning Objectives: Non-covalent molecular interactions

1. To have a broad introduction to the role of non-covalent interactions between molecules and their essential role in determining the properties of condensed-phase materials.
2. To recall the concept of electronegativity and bond polarity, then further to understand how molecular liquids have differing molecular **polarity**. The relationship between solubility and solvent polarity and polarisability.
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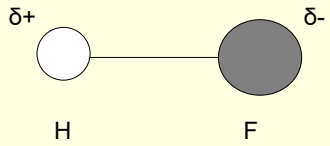
The '*London dispersion*', '*dipole-dipole*' and '*dipole-induced dipole*' interactions are generally all classed as **Van der Waals** interactions and are present between all molecules.

99

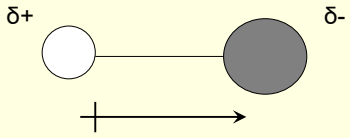
99

Van der Waals Interactions – dipole-dipole

If the electronic charge on a molecule is not evenly spread across the constituent atoms, then one side of the molecule may have a small positive charge and the other a small negative charge, e.g. HF



Here we can see that the F atom has acquired a negative charge relative to the H atom. We represent the polarity of this bond by using a dipole arrow:



The arrow head points to the negative end of the bond and the cross is found at the positive end.

100

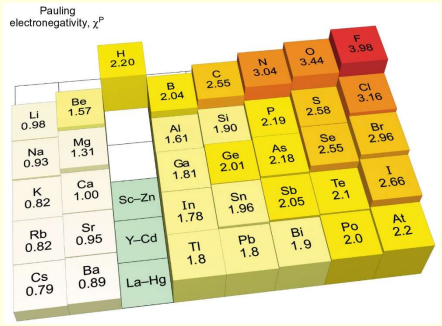
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The polarity of this bond is due to the differing **electronegativity** of the constituent atoms.

The 'shared' electrons in a covalent bond (or in a molecular orbital) are often not shared equally between the atoms.

The more electronegative atom attracts the electrons more strongly. The bond is then said to be 'polar', with greater electron density at one end than the other.

The Pauling Electronegativity scale from 0.0 – 4.0 is used to classify the electronegativity, χ, of each atom.



F, O, N, Cl, are the most electronegative atoms.

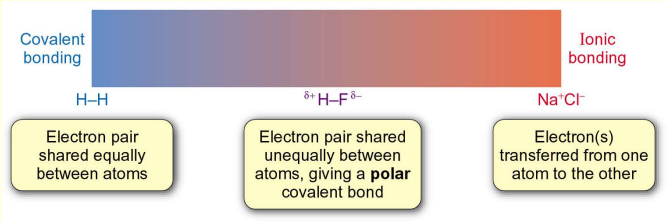
Electronegativity *increases* across period as the number of protons in the nucleus increases, giving a greater nuclear charge.

It *decreases* down the group as outer electrons get further from nucleus

101

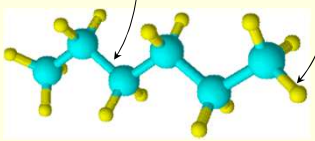
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The **polarity of a bond** can be (roughly) estimated by the difference between the χ of the two atoms. For $\chi^a - \chi^b > 2$, the bond is so 'polarised' that it can be considered as **ionic** (i.e. total transfer of electrons from one atom to the other).



C-C bonds in Hexane
$\Delta\chi = 2.55 - 2.55 = 0$

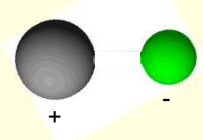
Non-polar covalent bonds



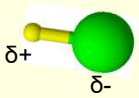
C-H bond in hexane
$\Delta\chi = 2.55 - 2.20 = 0.35$

Very weakly polar covalent bond

Na-Cl bond:
$\Delta\chi = 3.16 - 0.93 = 2.23$
Ionic !



H-Cl bond:
$\Delta\chi = 3.16 - 2.20 = 0.96$
Polar covalent bond



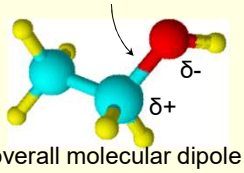
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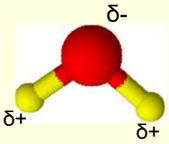
If the molecule contains more than two atoms, then the polarity of a bond cannot necessarily be considered in isolation. So in ethanol the O is also bonded to an H and the C is bonded to H and CH_3 groups. These also induce polarity into the bond.

C-O bond in ethanol
$\Delta\chi = 3.44 - 2.55 = 0.89$

Polar covalent bond



If more than one bond in a molecule is polar, then the overall molecular dipole moment (μ) of the molecule must be considered rather than simply the bond dipole, e.g. water:

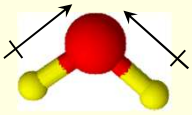


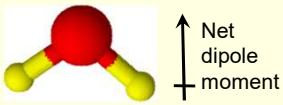
Oxygen: $\chi = 3.44$

Hydrogen: $\chi = 2.2$

O-H bonds are 'polarised' with more electron density around the Oxygen.

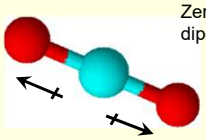
We represent the bond polarity using **dipole** arrows: arrow head at the negative end, cross at the positive end:





The overall molecular **dipole**, the sum of the individual bond dipole vectors, can be represented with a similar arrow. This is the **dipole moment, *μ*** .

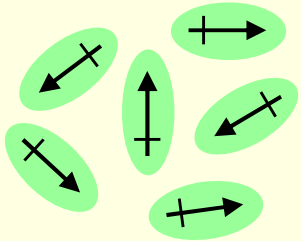
Molecular symmetry is an important factor in the polarity of a molecule. For example, the C=O bonds in CO_2 are polar, but CO_2 as a whole is non-polar because the individual bond dipoles cancel.



Zero net dipole moment

The **dipoles** of two neighbouring molecules can **interact** with each another, and overall the result is for the molecules to be **attracted together.** This dipole-dipole attraction is a type of van der Waals interaction.

Dipole – Dipole Interactions

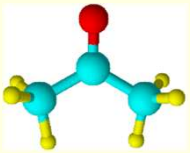


The overall average interaction energy between freely rotating dipolar molecules is dependent on the distance between them, *r*, to the sixth power:

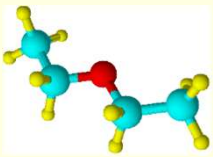
$$E \sim -\frac{1}{r^6}$$

It is also inversely dependent on the temperature, so the interaction becomes weaker as *T* increases.

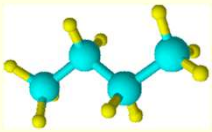
Dipole interactions could be used to rationalise the following data:



Acetone
Dipole moment, μ:
2.88 *D*
Boiling point (at 1 Atm):
56.2° C



Diethyl ether
Dipole moment, μ:
1.15 *D*
Boiling point (at 1 Atm):
34.5° C



Butane
Dipole moment, μ:
0 *D*
Boiling point (at 1 Atm):
-0.5° C

But how can we explain the interactions between non-polar molecules or atoms? For example the trend in boiling point of the noble gases suggests that the interaction between the atoms changes as we go down the group:



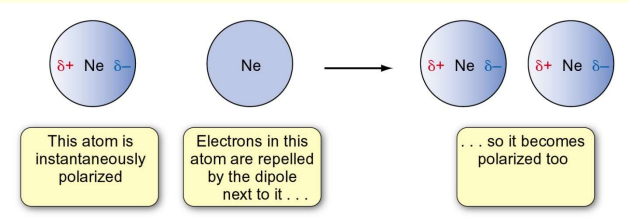
Boiling point (at 1 Atm), in **Kelvin**:

He	Ne	Ar	Kr	Xe
4.22	27.1	87.3	121	166 **K**

Van der Waals Interactions – dispersion interactions

Dipole-Induced-Dipole Interactions So how are non-polar molecules 'held together' if they don't have a dipole? The electrons in a molecule can be **polarised** due to the influence of a nearby polar molecule. The permanent dipole and **induced dipole** can then attract.

London Dispersion Interactions Dipoles can arise as a result of fluctuating electron density. This **instantaneous dipole** in a non-polar molecule can then induce a dipole in the neighbouring non-polar molecule. The **induced-dipoles attract**. This is called the '**Dispersion Interaction'**, and also varies as **$1/r^6$**.



Thus all molecules are attracted to all others through these different types of Van der Waals interactions. The size of the permanent dipole, or the **polarisability** of a non-polar molecule, determines how much attraction molecules feel for each other.

Example: **Chlorine** and **Bromine**

Cl_2 is a gas at room temperature and pressure. **Br_2** is a liquid under the same conditions. WHY?



Cl—Cl *Br—Br*

Neither Cl_2 or Br_2 are polar, so there cannot be a difference in permanent dipole interactions.



Consider the electronic structure of Cl and Br:
Chlorine: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$
Bromine: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ \mathbf{4s^2\ 3d^{10}\ 4p^5}$

More electron-rich Br_2 is more **polarisable** and therefore has much stronger **dispersion interactions** with neighbouring Br_2 molecules

Importance of non-covalent interactions in solvation

The polarity of a solvent is often discussed with respect to its **dielectric constant** also called the **relative permittivity**, ε_r.

Dielectric constants are measured experimental values that determine how well a material stores electrical energy when exposed to an electric field. Generally, more polar molecules have higher dielectric constants as they move in response to a field so that their negative and positive ends are correctly aligned.

While **dipole moment** is a property of an **individual molecule**, the **dielectric constant** is a property of the **bulk substance** (e.g. the liquid).

Notice that higher dipole moments generally lead to high dielectric constants.

		μ / D	ε_r
Benzene	C_6H_6	0.00	2.27
Carbon tetrachloride	CCl_4	0.00	2.23
Toluene	$C_6H_5CH_3$	0.36	2.40
Diethyl ether	$(C_2H_5)_2O$	1.15	4.34
Chloroform	$CHCl_3$	1.01	4.80
Ammonia	NH_3	1.47	16.90
Acetone	$(CH_3)_2C{=}O$	2.88	20.70
Ethanol	C_2H_5OH	1.69	24.30
Water	H_2O	1.85	78.54

Solvent properties are determined by a liquid's dielectric constant and polarity: **polar solvents solvate polar solutes; and non-polar solvents solvate non-polar solutes!**

Different types of non-covalent interactions

Table 1.10 Types of non-covalent interactions. For comparison, the energies of covalent bonds lie in the range 150 kJ mol^{-1} to 1000 kJ mol^{-1}.

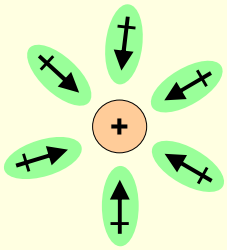
Interaction	Acts between	For example	Typical energy / kJ mol^{-1}
London dispersion	All types of molecules	H_2, CH_4, Cl_2, HCl	~ 5
Dipole–dipole	Polar molecules	HCl	2 (in gas phase)
Dipole–induced dipole	A polar molecule and a molecule that may or may not be polar	Between HCl and Cl_2 or between HCl and CH_4	< 2
Ion–dipole	An ion and a polar molecule	A metal ion dissolved in water	15
Hydrogen bond	A molecule containing the electronegative atoms O, N, or F bound to H and a molecule containing O, N, or F. The H atom forms a link between the two electronegative atoms	H_2O, NH_3, HF	10–40*
Ion–ion	Ionic groups	Between $-NH_3^+$ and $-CO_2^-$ groups in proteins	250

* The hydrogen bond in the ion, $[HF_2]^-$, is particularly strong, 165 kJ mol^{-1}; see Section 25.3 (p.1147).

Ion-dipole interactions and hydration of ions by water

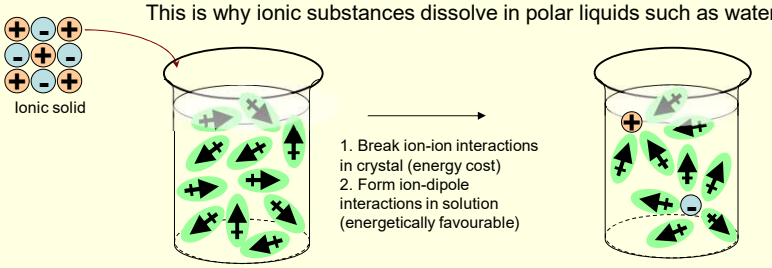
Dipoles do not only interact with one another: they also play a crucial role in **stabilising ions**.

When an ion is surrounded by polar molecules, many of their **dipoles align** in the field of the ion and this lowers the self-energy (energy that a particle has as a result of changes that it causes in its environment) of the ion.



If the solvent is water, this is call **hydration**. Hydration energies can be very large: e.g. 200 – 2000 kJ/mol at room temperature.

This is why ionic substances dissolve in polar liquids such as water.



Dipoles and polarity also influence the miscibility of liquids:

A rule of thumb: "*like dissolves like*"

"*like…*"
Where *like* means like polarity, or like hydrogen bonding, or like density

e.g. **solubility in water**:

Hexane	insoluble
Toluene	insoluble
Tetradecanoic acid	insoluble
Dichloromethane	slightly soluble
Butan-1-ol	soluble
Ethanol	very soluble
Glucose	very soluble
Urea	very soluble

e.g. **solubility in hexane**:

Water	insoluble
Tetradecanoic acid	very soluble
Toluene	very soluble

Structure, Bonding, and Self-Assembly of Biological Structures

CHEM0010/1 – Part A

Dr. Anna Roffey

Academic Support Hours:
Mon 11:30-12:30
Thurs 2-3 pm (CIB 124h)

Lecture 5



Learning Objectives: Non-covalent molecular interactions

1. To have a broad introduction to the role of non-covalent interactions between molecules and their essential role in determining the properties of condensed-phase materials.
2. To recall the concept of electronegativity and bond polarity, then further to understand how molecular liquids have differing molecular **polarity**. The relationship between solubility and solvent polarity and polarisability.
3. To focus on the interesting properties of water as nature's solvent, in particular the **dissociation** and **hydration** of ionic species in water.
4. To understand (qualitatively) the **hydrogen bonding** within bulk water, between water and solute molecules, and within and between other molecules.
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7. To encounter a range of **examples of biological self-assembly**.

Hydrogen bonding

To return to the table showing the relative energy of non-covalent interactions, we can see that hydrogen bonds are in general stronger than van der Waals interactions:

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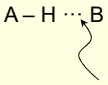
Interaction	Acts between	For example	Typical energy / kJ mol^{-1}	
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Dipole–dipole	Polar molecules	HCl	2	
Dipole–induced dipole	A polar molecule and a molecule that may or may not be polar	Between HCl and Cl_2 or between HCl and CH_4	< 2	
Ion–dipole	An ion and a polar molecule	A metal ion dissolved in water	15	
Hydrogen bond	A molecule containing the electronegative atoms O, N, or F bound to H and a molecule containing O, N, or F. The H atom forms a link between the two electronegative atoms	H_2O, NH_3, HF	10–40*	
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* The hydrogen bond in the ion, $[HF_2]^-$, is particularly strong, 165 kJ mol^{-1}; see Section 25.3 (p.1147).

While all molecules interact with all others through van der Waals interactions, **hydrogen bonds** are a special class of non-covalent interactions which only occur in particular circumstances.

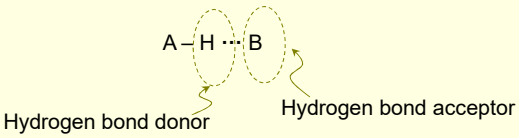
The 'text-book' definition of a hydrogen bond is:

The H-bond forms when <u>both</u> A and B are <u>strongly electronegative atoms</u> (N, O or F) and <u>B possesses a lone pair of electrons</u> with which to attract the H atom.



Hydrogen bond (represented by three dots)

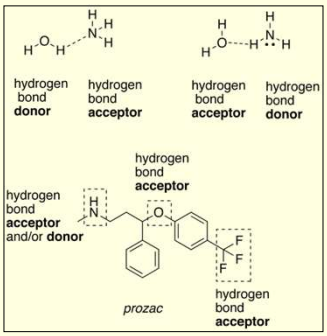
The H attached to an electronegative atom is called the '**hydrogen bond donor**'.
The electronegative atom which coordinates to this H is called the '**hydrogen bond acceptor**'.



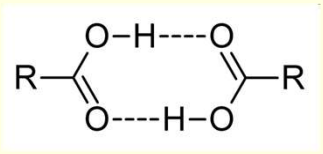
The formation of hydrogen bonds involves overlap of atomic orbitals on the A, H, and B atoms: therefore hydrogen bonds are very **short ranged** and A, H and B are usually linear (reducing electrostatic repulsion). **Hydrogen bonds are strong** compared to other non-covalent interactions (~20 kJ/mol) because of this orbital overlap.

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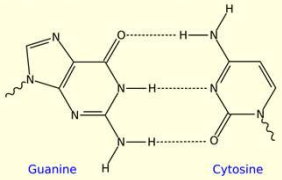


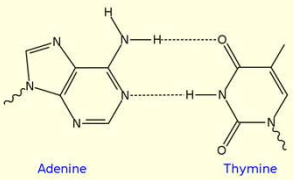
A molecule may be both a hydrogen bond donor and a hydrogen bond acceptor.



e.g. Dimerisation of carboxylic acids

Hydrogen bonds are one of the principal factors controlling the structure of biological macromolecules. The single strands of DNA are clasped into a double helix driven by strong hydrogen bonding.

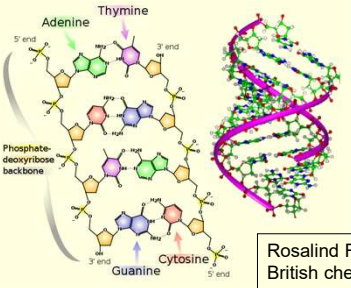




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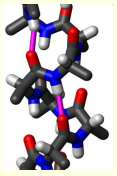
Its pretty amazing how such simple interactions at an atomic level allows the self-assembly of large macromolecules such as DNA. The relative strength of the bonds is also very important to its stability and function.

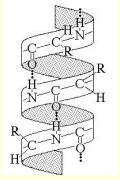


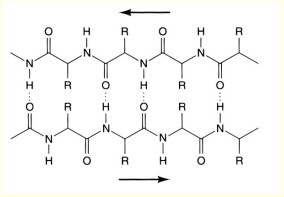


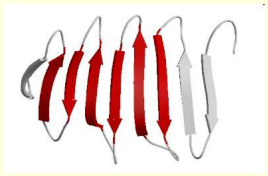
Rosalind Franklin (1920-1958)
British chemist and X-ray crystallographer – DNA, RNA, etc.

The same interactions are vital for determining the structure of proteins:









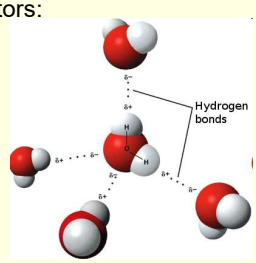
Both the α-helix and the β-sheet are caused by intramolecular hydrogen bonding. In fact, these are the optimal structures for maximising the hydrogen bonding within a poly-peptide molecule.

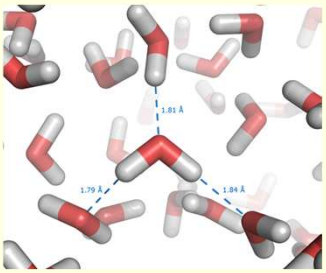
Water as nature's solvent: The importance of Hydrogen bonding



- Why does ice float on water?
- Why is the boiling point of H_2O 100°C, while that of that of H_2S is -85°C?
- Why is water such a good solvent?

The reasons for the anomalous properties of water is that each water molecule can form up to four hydrogen bonds with other water molecules – two as donors and two as acceptors:





The strong hydrogen-bonding network in water significantly increases the boiling point: in order to enter the gas phase, a molecule must break its hydrogen bonds, and this is energetically costly.

H_2O
Boiling point (at 1 Atm):
+100° C





H_2S
Boiling point (at 1 Atm):
-85° C

This also applies to other hydrogen bonding liquids such as ethanol or glycerine.

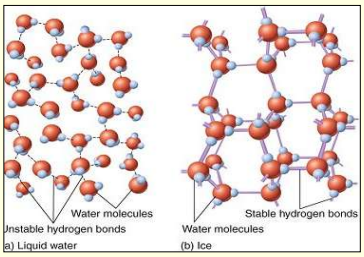


Glycerine (propane-1,2,3-triol)
Boiling point (at 1 Atm):
+290° C



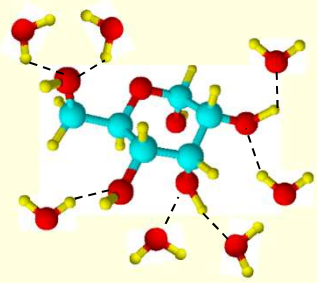
3-methylpentane
Boiling point (at 1 Atm):
+60° C

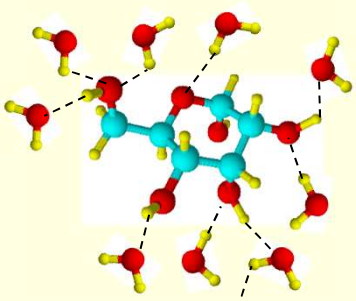
Usually during freezing molecules get closer together so solids are usually more dense than liquids. When water freezes, its molecules are held apart by strong, stable hydrogen bonds. This means that ice is less dense that water.



Solvation in water

Hydrogen bonding with surrounding water molecules strongly promotes solvation in water.

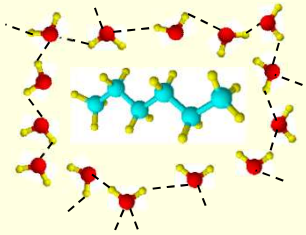




However, the H-bonding network is not 'frozen' in one orientation; water molecules can translate or rotate, breaking H-bonds and forming new ones. The lifetime of a typical hydrogen bond is ~ 1-10 *ps* (~ 10^{-12} -10^{-11} *s*).

Since there are ***many ways of arranging*** the hydrogen bonding network, each of the same energy, this leads to a favourable ***high entropy*** of the system. (And therefore, a low *free energy*).

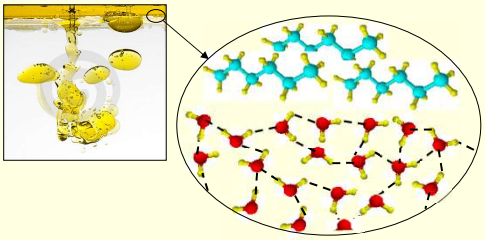
But what happens when we try to dissolve non-polar species in water? Hydrogen bonding cannot occur between water and a dissolved non-polar molecule. We call the non-polar molecule '***hydrophobic***'.



There are now ***fewer ways of arranging the H-bond network***, because those molecules near the non-polar molecule have fewer 'options' for H-bonding.

This ***reduces the entropy**** of the system, increasing the free energy.

The result: it is unfavourable for non-polar molecules to be surrounded by water, so they will tend to separate from water to minimise the contact area.



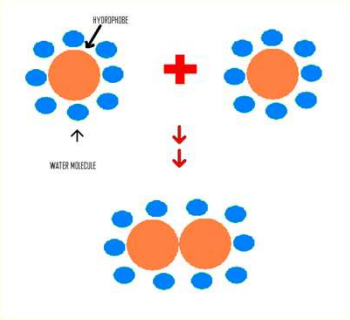
* You will learn about entropy, enthalpy, and free energy, and how they are related, in **Section D** of 001x

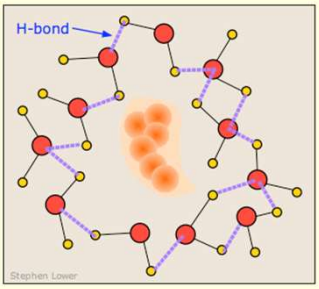
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7. To encounter a range of **examples of biological self-assembly**.

This collecting together of non-polar molecules is called ***the hydrophobic effect.***

It is also sometimes called the 'hydrophobic interaction', but this can be confusing because it suggests that it is caused by an attraction between the non-polar (hydrophobic) molecules. This is not the case: it is driven by a reduction in their (entropically) unfavourable interaction with water.





Some molecules contain polar and non-polar parts and are called **amphiphiles.**

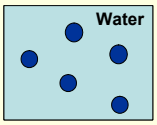
Amphiphile: from the Greek: αμφις, amphis: ***both*** and φιλία, philia: ***love***.



An amphiphile is a molecule which contains both water-loving and oil-loving parts.
The water-loving part is either charged (ionic) or polar or hydrogen bonding.
The oil-loving part is non-polar, e.g. hydrocarbon.

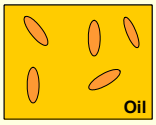


Polar or charged 'head' is favourably solvated in water

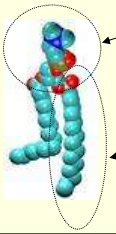




Hydrocarbon non-polar 'tail' is solvated in non-polar solvents, but not in water



An important example of an amphiphile in biology is a lipid:



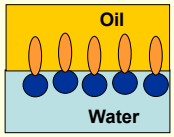
Typically lipids have a charged or polar headgroup (sometimes two or three charges per molecule).
The headgroups tend to be hydrated by surrounding water.

Lipids have one or more hydrocarbon tails, which are non-polar and therefore are not favourably solvated in water.

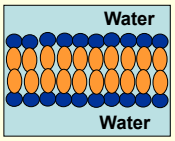
Amphiphile self-assembly

Amphiphiles ***self-assemble*** in order to minimise the (entropically) unfavourable interface between non-polar tails and the surrounding water.

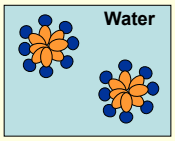
For example, amphiphiles **self-assemble** at interfaces, e.g. water-oil interface, to remove the non-polar tails from the water.



In a 'bath' of water, amphiphiles **self-assemble** to minimise hydrophobe-water surface area: the **hydrophobic effect.**



Self-assembled **bilayer.**



Self-assembled **micelles**

An important and illustrative example of self-assembly in biology is the packing of lipids into membranes. Membrane formation allows the non-polar 'tails' to be sequestered away from water while the headgroups are all hydrated.

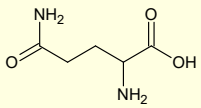
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Protein folding is self-assembly

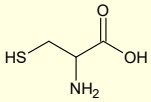
The folding of long chains of amino acids into functioning proteins occurs spontaneously, and is driven by **many different non-covalent interactions acting together**: hydrophobic effect, dipole-dipole interactions, ion-dipole interactions, dispersion interactions, and hydrogen bonding.

Hydrogen bonding side chains, e.g.



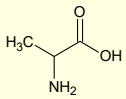
glutamine

Polar but non-hydrogen bonding side chains, e.g.

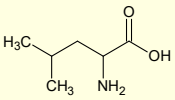


cysteine

Non-polar side chains, e.g.



alanine



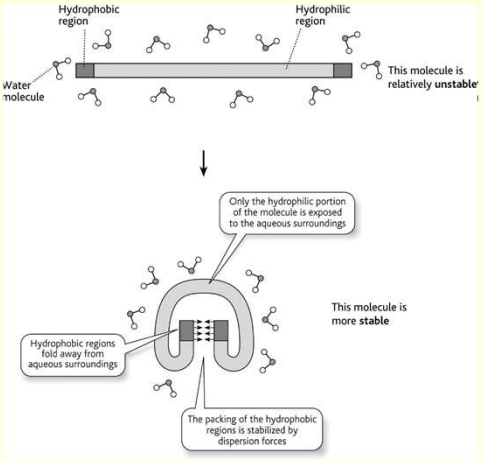
leucine

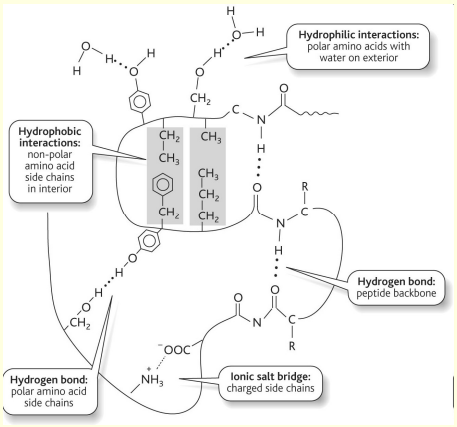
Proteins fold in such a way that the hydrophobic side chains are brought together, minimising their contact surface area with water and other hydrophilic groups. Hydrophilic side chains are exposed to water and the surrounding polar environment. Hydrogen bonding stabilises secondary structural elements β-pleated sheets and alpha helices.

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Protein folding is self-assembly





Want more?

This material is covered in Chemistry[3] – use the index to find the relevant sections

For more information see:

Richard A. L. Jones – "Soft Condensed Matter" – OUP 2002

Philip Nelson – "Biological Physics: Energy Information and Life" – WH Freeman and Co. NY 2004

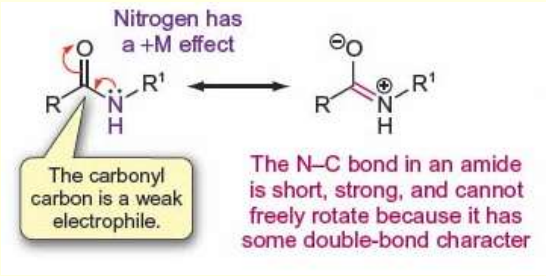
Philip Ball – "H_2O: A Biography of Water" – Weidenfeld & Nicholson 1999

For the latest definitions and discussions of the hydrogen bond see:

Definition of the hydrogen bond (IUPAC Recommendations 2011)

Pure and Applied Chemistry, Vol 83, 8, 1637 – 1641

Peptide Resonance Forms – why is N sp^2?



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